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dLGN Local GABAergic Interneurons: Structure, Development, Connectivity, Markers, and Heterogeneity

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Background

dLGN local interneurons are one of the two main inhibitory systems that shape visual signals in the geniculate nucleus. They are local GABAergic cells that help control how retinal input is passed on to thalamocortical relay neurons. (7 sources)

The dorsal lateral geniculate nucleus (dLGN) is the main thalamic relay for vision: relay neurons receive strong glutamatergic input from retinal ganglion cells and send visual signals to cortex, but they do not simply pass this input through unchanged. Instead, retinogeniculate transmission is strongly shaped by a much larger set of nonretinal inputs, including corticothalamic, brainstem, pretectal, thalamic reticular, and local interneuron inputs, with nonretinal terminals making up most synapses in dLGN. (Hammer et al., 2014) (Hammer et al._1, 2014) (Bickford et al., 2010)

Within this broader inhibitory control, dLGN local interneurons form the feedforward inhibitory system of the geniculate, in contrast to the recurrent inhibitory system centered in the perigeniculate/thalamic reticular region. Both inhibitory systems use GABA, but local interneurons are distinguished by their position within the main dLGN layers and by their role in shaping incoming visual signals locally. (Lindstrom et al., 2010)

A core anatomical feature of this inhibitory circuitry is the presence of GABAergic “F” terminals, which contain flattened, oval synaptic vesicles. In dLGN these inhibitory terminals come from two sources: thalamic reticular/perigeniculate neurons and local dLGN interneurons. This places local interneurons as intrinsic inhibitory elements embedded directly in the visual thalamic relay circuit, rather than as a distant modulatory input. (Hammer et al., 2014) (Hammer et al._1, 2014)

More broadly, the LGN fits the general pattern seen in brain regions that rely on GABAergic interneurons for signal modulation. In the dLGN, local interneurons are thus best viewed as a key part of the inhibitory machinery that adjusts the gain, timing, and fidelity of visual transmission before signals reach cortex. (Gamlin et al., 2018) (Gamlin et al._1, 2018) (Halassa et al., 2016)

Evidence

(Hammer et al., 2014)

Nuclei-specific differences in nerve terminal distribution, morphology, and development in mouse visual thalamus

"The overwhelming majority of these studies have focused on inputs to and projections from the dorsal lateral geniculate nucleus (dLGN). Relay neurons within dLGN receive strong glutamatergic inputs from retinal ganglion cells (RGCs) and serve as the principle conduit of visual signaling to the cortex. However, relay neurons do not act as passive relays of visual information. The gain of retinogeniculate signal transmission is modulated by nonretinal inputs to dLGN. These nonretinal inputs arise from visual cortex, pretectum, brainstem, thalamic reticular nuclei, and local dLGN interneurons, and they far outnumber the more powerful retinal inputs (Sherman et al., 2002)(Sherman, 2004). In fact, nonretinal inputs account for as much as 95% of the nerve terminals in dLGN (Sherman et al., 2002)(Erisir et al., 1997)(Erisir et al., 1997)(Horn et al., 2000)(Bickford et al., 2010)"

".One set of nerve terminals contain flattened, oval shaped synaptic vesicles, a hallmark feature of inhibitory GABAergic terminals, and are therefore called F terminals (Sherman, 2004)(Lund et al., 1972)(Famiglietti, 1970). In dLGN, GABAergic terminals arise from the thalamic reticular nucleus and local inhibitory interneurons [7,18]19]."

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Synaptic Development of the Mouse Dorsal Lateral Geniculate Nucleus

"The dorsal lateral geniculate nucleus (dLGN) of the mouse has emerged as a model system in the study of thalamic circuit development. However, there is still a lack of information regarding how and when various types of retinal and nonretinal synapses develop. We examined the synaptic organization of the developing mouse dLGN in the common pigmented C57/BL6 strain, by recording the synaptic responses evoked by electrical stimulation of optic tract axons, and by investigating the ultrastructure of identified synapses. At early postnatal ages (P14), when optic tract stimulation routinely evoked an excitatory postsynaptic potential/inhibitory postsynaptic potential (EPSP/IPSP) sequence, with the latter having both a GABAA and GABAB component. Electrophysiological and ultrastructural observations were consistent. At P7, many synapses were present, but synaptic profiles lacked the ultrastructural features characteristic of the adult dLGN, and little γ-aminobutyric acid (GABA) could be detected by using immunocytochemical techniques. In contrast, by P14, GABA staining was robust, mature synaptic profiles of retinal and nonretinal origin were easily distinguished, and the size and proportion of synaptic contacts were similar to those of the adult. The emergence of nonretinal synapses coincides with pruning of retinogeniculate connections, and the transition of retinal activity from spontaneous to visually driven. These results indicate that the synaptic architecture of the mouse dLGN is similar to that of other higher mammals, and thus provides further support for its use as a model system for visual system development. J. Comp. Neurol. 518:622–635, 2010. © 2009 Wiley-Liss, Inc."

(Lindstrom et al., 2010)

Feedforward and recurrent inhibitory receptive fields of principal cells in the cat's dorsal lateral geniculate nucleus

"Today, it is known that geniculate principal cells are equipped with two inhibitory circuits: a feedforward inhibitory system with the responsible interneurons located within the main layers of the dLGN and a recurrent inhibitory system with the interneurons confined to the perigeniculate nucleus [18,32]48]. Both types of inhibitory neurons have GABA as their main transmitter. The inhibitory circuits are similar but functionally segregated for X and Y cells (Lindström et al., 1990)."

(Gamlin et al., 2018)

Assembly and maintenance of GABAergic and Glycinergic circuits in the mammalian nervous system

"Brain regions such as the cortex, hypothalamus and lateral geniculate nucleus (LGN) within the thalamus primarily use GABAergic interneurons for signal modulation (reviewed by (Huang et al., 2007)(Decavel et al., 1990)(Halassa et al., 2016))."

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Thalamic inhibition: diverse sources, diverse scales

"The thalamus is the major source of cortical inputs shaping sensation, action and cognition. Thalamic circuits are targeted by two major inhibitory systems: the thalamic reticular nucleus (TRN) and extra-thalamic inhibitory (ETI) inputs. A unifying framework of how these systems operate is currently lacking. Here, we propose that TRN circuits are specialized to exert thalamic control at different spatiotemporal scales. Local inhibition of thalamic spike rates prevails during attentional selection whereas global inhibition more likely during sleep. In contrast, the ETI (arising from basal ganglia, zona incerta, anterior pretectum and pontine reticular formation) provides temporally-precise and focal inhibition, impacting spike timing. Together, these inhibitory systems allow graded control of thalamic output, enabling thalamocortical operations to dynamically match ongoing behavioral demands."

Structure, location, and heterogeneity

dLGN local interneurons are sparse inhibitory neurons embedded within the main relay nucleus rather than in a separate shell. In rodents they are often treated as a largely uniform class, but there is evidence for limited heterogeneity in size, physiology, and some molecular features. (10 sources)

In rodents, dLGN interneurons make up a minority of cells within the nucleus, with estimates ranging from about 10–20% in reviews and physiological studies, though unbiased stereology in adult mouse gave a lower estimate of about 5.6% of all dLGN neurons. They are one of the few substantial local inhibitory populations in dorsal thalamus, and unlike neighboring long-range inhibitory systems they form only local inhibitory networks within dLGN itself. (Su et al., 2020) (Evangelio et al., 2018) (Weil et al., 2022)

Their somata are distributed throughout the dLGN rather than being confined to a single lamina or border zone, and in mouse GFP-labeled interneurons were described as evenly distributed across the nucleus. Morphologically, they have highly complex, widespread dendritic arbors that can cross eye-specific borders, fitting with the idea that a single interneuron can sample and influence activity across broad parts of dLGN. At ultrastructural and circuit levels, this broad reach is paired with very local specializations: one reconstructed mouse interneuron innervated hundreds of thalamocortical cells across functionally distinct regions, mostly with weak en passant contacts, but also formed stronger multisynaptic relationships through specialized neurites. Notably, most of its neurites were “bifunctional,” both receiving retinal input and making inhibitory output synapses, reinforcing the unusual dendrite-like/axon-like mixed structure of these cells. (Seabrook et al., 2013) (Seabrook et al._1, 2013) (Morgan et al., 2019)

A distinctive structural feature is that these interneurons release GABA from both axons and dendrites, and their dendrites are electrically active. In rodent dLGN, dendrites can support brief sodium spikes and longer calcium spikes, whereas axons support sodium spikes only; this means dendritic branches are not passive receivers but active output compartments. This helps explain why the same cell can support both spatially broad and very local inhibition. (Antal et al., 2010)

At the class level, the rodent literature often describes dLGN interneurons as unusually homogeneous compared with neocortical interneurons. Several reviews and experimental papers state that there is just one, or at most one to two, interneuron types in rodent dLGN when judged by morphology, intrinsic membrane properties, gene expression, and developmental origin. (Antal et al., 2010) (Su et al., 2020)

Still, this apparent homogeneity is not absolute. Electrophysiological and molecular clustering has identified two interneuron subclasses in mouse dLGN: a small-cell class with more depolarized resting potential, stronger anomalous rectification, faster and higher-frequency firing, rebound bursting, and higher Ih density, and a large-cell class with lower Ih-related excitability. In that study, expression of neuronal nitric oxide synthase (nNOS) was detected only in the larger-diameter interneurons, suggesting at least one molecularly and physiologically distinct subtype. More broadly, somatic current injection studies and modeling also show a range of firing patterns across dLGN interneurons, including tonic firing, periodic bursting, burst-plus-tonic patterns, spike-frequency adaptation, and rebound bursts after hyperpolarization. (Leist et al., 2016) (Halnes et al., 2011)

Species differences also matter. In ferret LGN, a subset of interneurons in the interlaminar zones has been described as “PGN-like”: these cells resemble perigeniculate neurons electrophysiologically, pharmacologically, anatomically, and immunocytochemically, innervate restricted regions of the A laminae,

and are parvalbumin-positive. This suggests that heterogeneity may be more obvious in laminated carnivore LGN than in rodent dLGN, where local interneurons are more often grouped into a single broad class. (Sanchez-Vives et al., 1996)

Evidence

(Su et al., 2020)

Retinal inputs signal astrocytes to recruit interneurons into visual thalamus

"In rodents, the primary retinorecipient regions of visual thalamus include the dorsal lateral geniculate nucleus (dLGN), ventral LGN (vLGN), and intergeniculate leaflet (IGL). These nuclei receive input from a highly diverse group of retinal ganglion cells (RGCs) but only the dLGN serves as a relay of retinal information to the visual cortex (4)(5)(6). Unlike many other regions of rodent thalamus, these three nuclei each contain local GABAergic interneurons (6)(7)(8).

The dLGN, similar to the neocortex, contains a small percentage (10 to 20%) of GABAergic interneurons (8)(9)(10). In contrast to the great diversity of interneuron subtypes in the neocortex (1), dLGN interneurons are homogeneous, clustering into just one or two subtypes based on morphology, intrinsic membrane properties, gene expression, and embryonic origin (11)(12)(13)(14). GABAergic interneurons in dLGN receive monosynaptic input from RGCs and provide feedforward inhibition onto excitatory thalamocortical relay cells (15)(16)(17). Such input serves to sharpen receptive field properties and improve the contrast-gain of retinogeniculate transmission (17)(18)(19)(20)."

(Evangelio et al., 2018)

Thalamocortical Projection Neuron and Interneuron Numbers in the Visual Thalamic Nuclei of the Adult C57BL/6 Mouse

"A key parameter to constrain predictive, bottom-up circuit models of a given brain domain is the number and position of the neuronal populations involved. These include not only the neurons whose bodies reside within the domain, but also the neurons in distant regions that innervate the domain. The mouse visual cortex receives its main subcortical input from the dorsal lateral geniculate nucleus (dLGN) and the lateral posterior (LP) complex of the thalamus. The latter consists of three different nuclei: lateral posterior lateral (LPL), lateral posterior medial rostral (LPMR), and lateral posterior medial caudal (LPMC), each exhibiting specific patterns of connections with the various visual cortical areas. Here, we have determined the number of thalamocortical projection neurons and interneurons in the LP complex and dLGN of the adult C57BL/6 male mouse. We combined Nissl staining and histochemical and immunolabeling methods for consistently delineating nuclei borders, and applied unbiased stereological cell counting methods. Thalamic interneurons were identified using GABA immunolabeling. The C57BL/6 dLGN contains ≈ 21,200 neurons, while LP complex contains ≈ 31,000 total neurons. The dLGN and LP are the only nuclei of the mouse dorsal thalamus containing substantial numbers GABA-immunoreactive interneurons. These interneurons, however, are scarcer than previously estimated; they are 5.6% of dLGN neurons and just 1.9% of the LP neurons. It can be thus inferred that the dLGN contains ≈ 20,000 and the LP complex ≈ 30,400 thalamocortical projection neurons (≈ 12,000 in LPL, 15,200 in LPMR, and 4,200 in LPMC). The present dataset is relevant for constraining models of mouse visual thalamocortical circuits, as well as for quantitative comparisons between genetically modified mouse strains, or across species."

(Weil et al., 2022)

Daily changes in light influence mood via inhibitory networks within the thalamic perihabenular nucleus

"Unlike the majority of the dorsal thalamus in mice, the retino-recipient dorsal lateral geniculate nucleus (dLGN) contains a subset of inhibitory (GABAergic) interneurons that spatially and temporally synchronize local networks (15)(Halassa et al., 2019)(Sherman, 2004)"

".These long-range inhibitory connections differentiate the PHb from the dLGN, which only contains local inhibitory networks (Morgan et al., 2019)."

(Seabrook et al., 2013)

Interneurons in the mouse visual thalamus maintain a high degree of retinal convergence throughout postnatal development

"Far less is known about the pattern of retinal convergence onto interneurons, the other principal cell type present in dLGN. These local-circuit neurons are involved in feedforward inhibition onto relay cells and play a role in contrast gain control, shaping receptive fields of relay cells, and altering the temporal precision of retinal inputs (Sherman, 2004)(Wang et al., 2011)"

".Here we overcame these obstacles by using transgenic mice that express enhanced green fluorescent protein (GFP) in γ-aminobutyric acid (GABA)ergic interneurons (GAD67-GFP) (Chattopadhyaya et al., 2004). Such cell-type specific visualization via GFP allowed us to readily target interneurons for in vitro recordings"

"GFP-expressing interneurons were evenly distributed throughout dLGN and had highly complex and widespread dendritic processes that often crossed eye-specific borders. Estimates of retinal convergence derived from excitatory postsynaptic potential (EPSP)

amplitude by stimulus intensity plots revealed that unlike relay cells, interneurons recorded throughout the first 5 weeks of life, maintain a large number (approximately eight to ten) of retinal inputs"

".The lack of pruning onto interneurons suggests that the activity-dependent refinement of retinal connections in dLGN is cell-type specific. The high degree of retinal convergence onto interneurons may be necessary for these cells to provide both widespread and local forms of inhibition in dLGN."

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An individual interneuron participates in many kinds of inhibition and innervates much of the mouse visual thalamus

"One way to assess a neuron's function is to describe all its inputs and outputs. With this goal in mind, we used serial section electron microscopy to map 899 synaptic inputs and 623 outputs in one inhibitory interneuron in a large volume of the mouse visual thalamus. This neuron innervated 256 thalamocortical cells spread across functionally distinct subregions of the visual thalamus. All but one of its neurites were bifunctional, innervating thalamocortical and local interneurons while also receiving synapses from the retina. We observed a wide variety of local synaptic motifs. While this neuron innervated many cells weakly, with single en passant synapses, it also deployed specialized branches that climbed along other dendrites to form strong multi-synaptic connections with a subset of partners. This neuron's diverse range of synaptic relationships allows it to participate in a mix of global and local processing but defies assigning it a single circuit function."

(Antal et al., 2010)

Cholinergic Activation of M2 Receptors Leads to Context-Dependent Modulation of Feedforward Inhibition in the Visual Thalamus

"In the rodent dLGN, there appears to be just one type of interneuron (Acuña-Goycolea et al., 2008)[16](Williams et al., 1996). The dLGN is a vital link in the visual pathway from the eye to the visual cortex [16,18]. Retinal ganglion cells (RGCs) activate thalamocortical (TC) neurons that convey visually evoked signals to the cortex, and local interneurons within the dLGN that provide feedforward inhibition to TC cells. dLGN interneurons control the number of visually evoked spikes, regulate the precision of firing, and refine the receptive fields (RFs) of TC neurons (Berardi et al., 1984)(Blitz et al., 2005)(Usrey et al., 1999)(Sillito et al., 1983) (Norton et al., 1992)(Norton et al., 1989)"

".dLGN interneurons release GABA from both their axons and dendrites [16](Cox et al., 1998). Their dendrites support both brief sodium action potentials and longlasting calcium spikes, whereas their axons support only sodium spikes. Thus, active responses in the postsynaptic cell control the magnitude, spatial properties and time course of GABA release."

(Leist et al., 2016)

Two types of interneurons in the mouse lateral geniculate nucleus are characterized by different h-current density

"Although hyperpolarization-activated cyclic nucleotide-gated cation (HCN) channels and the corresponding h-current (Ih) have been shown to fundamentally shape the activity pattern in the thalamocortical network, little is known about their function in local circuit GABAergic interneurons (IN) of the dorsal part of the lateral geniculate nucleus (dLGN). By combining electrophysiological, molecular biological, immunohistochemical and cluster analysis, we characterized the properties of Ih and the expression profile of HCN channels in IN. Passive and active electrophysiological properties of IN differed. Two subclasses of IN were resolved by unsupervised cluster analysis. Small cells were characterized by depolarized resting membrane potentials (RMP), stronger anomalous rectification, higher firing frequency of faster action potentials (APs), appearance of rebound bursting, and higher Ih current density compared to the large IN. The depolarization exerted by sustained HCN channel activity facilitated neuronal firing. In addition to cyclic nucleotides, Ih in IN was modulated by PIP2 probably based on the abundant expression of the HCN3 isoform. Furthermore, only IN with larger cell diameters expressed neuronal nitric oxide synthase (nNOS). It is discussed that Ih in IN is modulated by neurotransmitters present in the thalamus and that the specific properties of Ih in these cells closely reflect their modulatory options."

(Halnes et al., 2011)

A Multi-Compartment Model for Interneurons in the Dorsal Lateral Geniculate Nucleus

> *"The dorsal lateral geniculate nucleus (dLGN) receives input from retinal ganglion cells and transmits processed information to visual cortex. About 75-80% of the neurons in the dLGN are thalamocortical (TC) neurons, also called relay neurons, as they relay information from the retina to the cortex. Local GABAergic interneurons (INs) constitute the remaining 20-25%, and are responsible for most of the intra-nuclear connections [1]. By providing feed-forward inhibition from retinal ganglion cells to TC neurons, INs control the number of visually evoked spikes in TC neurons, and refine the receptive fields of TC neurons (e.g. (Hamos et al., 1985) (Norton et al., 1992)(Sherman, 2004)(Blitz et al., 2005)). The INs are also important for synchronizing thalamic oscillations (Steriade et al., 1993)(Steriade et al., 1996)"*
>
> *".For dLGN, a host of physiological and anatomical experiments have provided detailed information about the neuronal connectivity, as well as the morphology and physiology of TC neurons and INs (see e.g. reviews in [22][23][24])"*
>
> *".We here present a detailed compartmental model of INs using, for the first time, a morphological reconstruction and a set of active dendritic conductances constrained by experimental somatic recordings from INs under several different current-clamp conditions"*
>
> *".The INs exhibit a rich variety of firing patterns: Depolarizing current injections to the soma may induce tonic firing, periodic bursting or an initial burst followed by tonic spiking, sometimes with prominent spike-time adaptation. When released from hyperpolarization, some INs elicit rebound bursts, while others return more passively to the resting potential."*

(Sanchez-Vives et al., 1996)

Are the Interlaminar Zones of the Ferret Dorsal Lateral Geniculate Nucleus Actually Part of the Perigeniculate Nucleus?

> *"The ferret dorsal lateral geniculate nucleus (LGNd) contains interneurons within the interlaminar zones situated between the laminae corresponding to the ipsi-and contralateral eyes. We found that a subset of these neurons exhibits electrophysiological properties similar to those previously reported for perigeniculate (PGN) neurons, including the generation of rhythmic sequences of rebound low-threshold Ca 2 spikes at a frequency of 1-4 Hz after the intracellular injection of a hyperpolarizing current pulse. These "PGN-like" interlaminar interneurons innervated restricted regions of the A-laminae, inhibited thalamocortical cells through GABA A , and perhaps GABA B , receptors, and were excited by axon collaterals from thalamocortical cells. This reciprocal relationship is identical to that formed by PGN cells and allowed the PGN-like interlaminar neurons to participate in the generation of spindle waves and other network oscillations. Pharmacologically, PGN-like interlaminar interneurons were also similar to PGN neurons: both generated a prolonged depolarization in response to the local application of serotonin, 1S,3R-ACPD, and CCK8S, and a rapid depolarization followed by a more prolonged hyperpolarization in response to acetylcholine"*
>
> *".Examination of parvalbumin and calbindin staining in the ferret LGNd revealed that both PGN and a subset of interlaminar neurons were parvalbumin-positive. In contrast, calbindinpositive cells were relatively absent in the PGN and sparsely present in the interlaminar zones, but were numerous in the A and C laminae. Our results indicate that the interlaminar zone in between laminae A and A1 and A1 and C in the ferret LGNd possesses a cell type that is electrophysiologically, pharmacologically, anatomically, immunocytochemically, and functionally similar to neurons in the PGN."*

Developmental origin and maturation

dLGN local interneurons have an unusual developmental origin for thalamic inhibitory neurons: they arise from Sox14/Otx2-expressing precursors in the dorsal midbrain and then migrate into visual thalamus. Their later maturation depends strongly on retinal input, which helps recruit them into dLGN and supports postnatal arbor growth, pruning, and establishment of mature inhibitory output. (5 sources)

Unlike reticular thalamic GABAergic neurons, dLGN local interneurons do not arise from a local diencephalic source. Instead, mouse studies show that they are generated from Sox14/Otx2-expressing precursor cells in the dorsal midbrain (tectal region) and then migrate into the dLGN during early postnatal development, giving them an unusual extra-diencephalic origin among thalamic inhibitory neurons. (Jager et al., 2016)

Work on visual thalamus development suggests that recruitment of these interneurons into dLGN uses signals linked to retinal innervation. A shared mechanism appears to operate in both dLGN and vLGN, in which retinal ganglion cell axons act through thalamic astrocytes to induce fibroblast growth factor 15 (FGF15), a motogen needed for GABAergic interneuron migration into visual thalamus. More specifically, retinal ganglion cells provide sonic hedgehog (SHH), and loss of RGC-derived SHH reduces thalamic Fgf15 expression and lowers the fraction of interneurons recruited into visual thalamus. (Somaiya et al., 2022)

After they reach and disperse within dLGN, interneurons continue through a multistage postnatal maturation

program. In mouse, migration and dispersion are followed by arbor remodeling that begins in the second postnatal week. This includes an initial phase of exuberant branch growth, with increasing arbor number, complexity, and field size, followed by pruning and stabilization as cells acquire the more characteristic bipolar morphology described in mature dLGN. (Charalambakis et al., 2019)

Retinal input appears to be important not only for bringing interneurons into visual thalamus, but also for shaping their later structural and synaptic maturation. In math5/Atoh7 mutant mice that lack normal retinal innervation, dLGN interneurons fail to undergo the usual branch growth and pruning sequence and instead retain a sparse, simple morphology. In the same study, activation of mGluR1/5-sensitive F2-related signaling with DHPG increased inhibitory synaptic activity in relay neurons in wild type but had little effect in math5-null mice, consistent with impaired maturation of interneuron output when retinal drive is missing. (Charalambakis et al., 2019)

Development of retinal input onto interneurons also differs from the well-known refinement seen in relay cells. During the first five postnatal weeks, relay neurons undergo activity-dependent pruning of retinal inputs, but interneurons retain a high degree of retinal convergence, on the order of about 8-10 retinal inputs per cell. This suggests that retinogeniculate refinement is cell-type specific, with interneurons preserving broad retinal sampling as they mature. (Seabrook et al., 2013)

Overall, current evidence supports a developmental sequence in which tectum-derived interneuron precursors are recruited into visual thalamus by retina-dependent signaling, then use ongoing retinal input to complete postnatal arbor remodeling and establish the mature local inhibitory circuitry of dLGN. At the same time, the field still has important gaps: compared with relay neurons, much less is known about the developmental logic of intrinsic interneurons, and even recent work notes that key roles of retinal input in related visual thalamic nuclei remain incompletely understood. (Charalambakis et al., 2019) (Seabrook et al., 2013) (Stebbins et al., 2024)

Evidence

(Jager et al., 2016)

Tectal-derived interneurons contribute to phasic and tonic inhibition in the visual thalamus

"The release of GABA from local interneurons in the dorsal lateral geniculate nucleus (dLGN-INs) provides inhibitory control during visual processing within the thalamus"

". during early postnatal development Sox14/Otx2-expressing precursor cells migrate from the dorsal midbrain to generate dLGN-INs. The unexpected extra-diencephalic origin of dLGN-INs sets them apart from GABAergic neurons of the reticular thalamic nucleus. Using optogenetics we show that at increased firing rates tectal-derived dLGN-INs generate a powerful form of tonic inhibition that regulates the gain of thalamic relay neurons through recruitment of extrasynaptic high-affinity GABAA receptors"

". Although interneurons are ubiquitous in the thalamus of carnivorans and primates, they are largely confined to the dorsal lateral geniculate nucleus (dLGN) in rodents 28 . Retinal input to the dLGN directly excites local GABAergic interneurons (dLGN-INs) [29][30][31] providing visually driven feed-forward inhibition of dLGN thalamic relay neurons 32 . The F2 terminals formed by dLGN-INs also display atypical wiring, releasing GABA at dendro-dendritic sites 33 found within triadic structures known as glomeruli 34 . In addition, F1 terminals are more classical axo-dendrtic synaptic arrangements."

(Somaiya et al., 2022)

Sonic hedgehog-dependent recruitment of GABAergic interneurons into the developing visual thalamus

"Despite these differences, there are some similarities in the cell types of dLGN and vLGN, including the presence and distribution of local interneurons (Arcelli et al., 1997)(Evangelio et al., 2018)(Jaubert-Miazza et al., 2005). In fact, a shared mechanism appears to exist that contributes to the recruitment and integration of these GABAergic neurons into both dLGN and vLGN (Golding et al., 2014;Su et al., 2020)"

".We previously reported that RGC axons signal astrocytes to induce the expression of fibroblast growth factor 15 (FGF15), a motogen required for GABAergic interneuron migration into visual thalamus"

".Specifically, we show that thalamus-projecting RGCs express SHH and thalamic astrocytes generate downstream components of SHH signaling. Deletion of RGC-derived SHH leads to a significant decrease in Fgf15 expression, as well as in the percentage of

interneurons recruited into visual thalamus."

(Charalambakis et al., 2019)

Developmental Remodeling of Thalamic Interneurons Requires Retinal Signaling

"little is known about the development of the other principal cell type, intrinsic interneurons"

"in dLGN interneurons"

"their development is a multistaged process involving migration, arbor remodeling, and synapse formation. Arbor remodeling begins during the second postnatal week, after migration to and dispersion within dLGN is complete. This phase includes a period of exuberant branching where arbors grow in number, complexity, and field size. Such growth is followed by branch pruning and stabilization, as interneurons adopt a bipolar architecture"

"To test how such defects influence connectivity with dLGN relay neurons, we used DHPG [(RS)-3,5-dihydroxyphenylglycine], the mGluR1,5 agonist that targets F2 terminals. This led to substantial increases in IPSC activity among WT relay neurons but had little impact in math5−/− mice"

"intrinsic interneurons"

"relies on retinal innervation"

"The absence of retinal signaling disrupts this process. The math5−/− interneurons fail to branch and prune, and instead maintain a simple, sparse architecture"

"GAD67-GFP mice, which express GFP in dLGN interneurons"

(Seabrook et al., 2013)

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"GFP-expressing interneurons were evenly distributed throughout dLGN and had highly complex and widespread dendritic processes that often crossed eye-specific borders. Estimates of retinal convergence derived from excitatory postsynaptic potential (EPSP) amplitude by stimulus intensity plots revealed that unlike relay cells, interneurons recorded throughout the first 5 weeks of life, maintain a large number (approximately eight to ten) of retinal inputs"

".The lack of pruning onto interneurons suggests that the activity-dependent refinement of retinal connections in dLGN is cell-type specific. The high degree of retinal convergence onto interneurons may be necessary for these cells to provide both widespread and local forms of inhibition in dLGN."

(Stebbins et al., 2024)

Retinal Input Is Required for the Maintenance of Neuronal Laminae in the Ventrolateral Geniculate Nucleus

"In contrast, little is known about the role of retinal inputs in the development of vLGN."

Connectivity and circuit organization

dLGN local interneurons sit directly in the retinogeniculate pathway. They receive strong retinal and corticothalamic drive, and in turn inhibit relay cells through both standard axonal synapses and unusual presynaptic dendrites organized into retinally driven triads. (15 sources)

Circuit-wise, dLGN interneurons are the intrinsic feedforward inhibitory nodes of the visual thalamus. Retinal ganglion cell (RGC) axons directly excite these GABAergic neurons, and interneurons then inhibit thalamocortical/relay neurons, helping place inhibition immediately next to incoming retinal excitation rather than only through external sources such as the thalamic reticular nucleus (TRN). This local inhibitory system operates alongside many nonretinal inputs to dLGN, but is distinct in being embedded within the relay nucleus itself. (Jager et al., 2016) (Yang et al., 2017) (Hammer et al., 2014) (Hammer et al._1, 2014)

A central organizational feature is that a single interneuron provides two forms of output onto relay cells. First, it makes conventional axodendritic GABAergic synapses through axonal boutons, classically called F1 terminals. Second, it makes dendrodendritic output through presynaptic dendrites, called F2 terminals. TRN neurons provide only the F1-like axonal form, whereas local dLGN interneurons provide both, making them unusual among thalamic inhibitory neurons. (Errington et al., 2011) (Yang et al., 2017) (Jager et al., 2016)

The best known microcircuit built from these dendritic outputs is the synaptic triad, often located in retinogeniculate glomeruli. In this arrangement, an RGC terminal contacts both a relay-cell dendrite and an interneuron dendritic appendage; that same interneuron dendrite then releases GABA back onto the nearby relay-cell dendrite. This lets one retinal input produce fast monosynaptic excitation and near-simultaneous disynaptic inhibition at very close range, a layout well suited for tight temporal control of relay-cell responses. (Errington et al., 2011) (Pressler et al., 2013) (Yang et al., 2017)

These two output compartments are not controlled in the same way. F1 release is action-potential dependent, whereas F2 dendritic release can occur locally and does not require a somatically generated spike. At F2 terminals, glutamate from retinal input can activate ionotropic and metabotropic glutamate receptors, especially mGluR5, and drive local calcium entry that triggers GABA release from the presynaptic dendrite itself. This supports the idea that many dendritic branches act as semi-independent input-output units rather than only as passive parts of a whole-cell integrator. (Errington et al., 2011) (Govindaiah et al., 2006) (Godwin et al., 1996) (Pressler et al., 2013) (Crandall et al., 2012)

At the same time, interneurons are not purely local branch processors. Action potentials initiated by somatic current injection or synaptic stimulation can backpropagate with high fidelity throughout the dendritic arbor and into the fine GABA-releasing appendages, producing near-synchronous GABA release from both axonal and dendritic compartments. This means dLGN interneurons can switch between branch-specific inhibition and cell-wide coordinated inhibition depending on how they are activated. (Casale et al., 2011)

Their input map is also broad. Compared with relay neurons, dLGN interneurons retain unusually high retinal convergence, around 8-10 retinal inputs per cell across early postnatal development, and their dendrites spread widely across the nucleus, often crossing eye-specific borders. This broad retinal sampling likely helps the same interneuron support both spatially widespread inhibition and highly local triadic inhibition. A recent connectomic/functional view goes further and suggests that each interneuron may globally encode a particular visual feature, much of it inherited from retinal input, and distribute feature-selective inhibition across many relay cells. (Seabrook et al._1, 2013) (Mullner et al., 2023)

dLGN interneurons are also recruited by nonretinal pathways. Corticothalamic axons from visual cortex densely innervate relay cells, TRN neurons, and local dLGN interneurons, so cortical feedback can excite interneurons directly and also generate disynaptic IPSPs in relay cells through local inhibition. Pharmacology and physiology further indicate that optic tract stimulation can enhance interneuron-mediated inhibition, whereas corticothalamic activity can regulate this inhibitory output through subtype-specific mGluR actions. (Krosigk et al., 1999) (Govindaiah et al., 2006)

This arrangement creates several nested inhibitory motifs. Retinal afferents recruit classic feedforward inhibition through local interneurons; corticothalamic afferents can both excite relay cells and recruit local disynaptic inhibition; and interneuron dendrites can provide ultralocal inhibition at individual retinal synapses while the same cell can also impose broader inhibition through backpropagation-driven release. At higher firing rates, interneurons can additionally generate a strong tonic inhibitory component onto relay neurons via extrasynaptic high-affinity GABA_A receptors, extending their influence beyond point-to-point synapses. (Jager et al., 2016) (Errington et al., 2011)

There are also species- and subtype-related differences in circuit embedding. In ferret, a subset of interlaminar interneurons is “PGN-like”: these cells are excited by thalamocortical collaterals, inhibit relay cells, and participate in reciprocal oscillatory loops similar to those of perigeniculate/TRN neurons, suggesting that in more laminated LGN the boundary between local interneuron and PGN-like circuit roles

can be less sharp than in rodents. A recent mouse study also reported that pyramidal-tract-type cortical input targets dLGN interneurons more strongly than relay cells, producing net disinhibition of relay responses to retinal input, which adds another pathway-specific layer to local circuit organization. (Sanchez-Vives et al., 1996) (Whitley et al., 2025)

Evidence

(Jager et al., 2016)

Tectal-derived interneurons contribute to phasic and tonic inhibition in the visual thalamus

"The release of GABA from local interneurons in the dorsal lateral geniculate nucleus (dLGN-INs) provides inhibitory control during visual processing within the thalamus"

". during early postnatal development Sox14/Otx2-expressing precursor cells migrate from the dorsal midbrain to generate dLGN-INs. The unexpected extra-diencephalic origin of dLGN-INs sets them apart from GABAergic neurons of the reticular thalamic nucleus. Using optogenetics we show that at increased firing rates tectal-derived dLGN-INs generate a powerful form of tonic inhibition that regulates the gain of thalamic relay neurons through recruitment of extrasynaptic high-affinity GABAA receptors"

". Although interneurons are ubiquitous in the thalamus of carnivorans and primates, they are largely confined to the dorsal lateral geniculate nucleus (dLGN) in rodents 28 . Retinal input to the dLGN directly excites local GABAergic interneurons (dLGN-INs) [29][30] [31] providing visually driven feed-forward inhibition of dLGN thalamic relay neurons 32 . The F2 terminals formed by dLGN-INs also display atypical wiring, releasing GABA at dendro-dendritic sites 33 found within triadic structures known as glomeruli 34 . In addition, F1 terminals are more classical axo-dendrtic synaptic arrangements."

(Yang et al., 2017)

Distinct kinetics of inhibitory currents in thalamocortical neurons that arise from dendritic or axonal origin

"Inhibitory inputs to thalamocortical neurons primarily arise from thalamic reticular nucleus (TRN) neurons and local dLGN interneurons. The local interneurons are activated by retinogeniculate afferents, providing feed-forward inhibition to thalamocortical neurons, and these inputs play a significant role in temporal precision and receptive field properties of thalamocortical neurons (Sillito et al., 1983)(Berardi et al., 1984)(Holdefer et al., 1989)(Blitz et al., 2005)"

".The dLGN interneurons are unique and interesting in that they give rise to two distinct types of output onto thalamocortical neurons: classical axonal outputs via F1 terminals and presynaptic dendrites (named F2 terminals) (Guillery, 2004)(Ralston, 1971) (Famiglietti et al., 1972)[18](Cox et al., 1998)(Crandall et al., 2012)(Crandall et al., 2013). Retinal inputs onto thalamocortical neurons provide monosynaptic excitation via a glutamatergic retinogeniculate synapse and disynaptic inhibition via the F2 terminal (presynaptic dendrite of interneuron). These two synapses onto the thalamocortical neurons are within very close proximity, and form a triadic structure [2]22]"

".Local GABAergic interneurons provide two distinct synaptic outputs: axonal (F1 terminals) and dendritic (F2 terminals) onto dLGN thalamocortical neurons. By contrast, TRN neurons provide only axonal output (F1 terminals) onto dLGN thalamocortical neurons."

(Hammer et al., 2014)

Nuclei-specific differences in nerve terminal distribution, morphology, and development in mouse visual thalamus

"The overwhelming majority of these studies have focused on inputs to and projections from the dorsal lateral geniculate nucleus (dLGN). Relay neurons within dLGN receive strong glutamatergic inputs from retinal ganglion cells (RGCs) and serve as the principle conduit of visual signaling to the cortex. However, relay neurons do not act as passive relays of visual information. The gain of retinogeniculate signal transmission is modulated by nonretinal inputs to dLGN. These nonretinal inputs arise from visual cortex, pretectum, brainstem, thalamic reticular nuclei, and local dLGN interneurons, and they far outnumber the more powerful retinal inputs (Sherman et al., 2002)(Sherman, 2004). In fact, nonretinal inputs account for as much as 95% of the nerve terminals in dLGN (Sherman et al., 2002)(Erisir et al., 1997)(Erisir et al., 1997)(Horn et al., 2000)(Bickford et al., 2010)"

".One set of nerve terminals contain flattened, oval shaped synaptic vesicles, a hallmark feature of inhibitory GABAergic terminals, and are therefore called F terminals (Sherman, 2004)(Lund et al., 1972)(Famiglietti, 1970). In dLGN, GABAergic terminals arise from the thalamic reticular nucleus and local inhibitory interneurons [7,18]19]."

(Hammer et al._1, 2014)

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(Errington et al., 2011)

mGluR control of interneuron output regulates feedforward tonic GABAA inhibition in the visual thalamus

"In the dorsal lateral geniculate nucleus (dLGN), interneurons can alter the temporal precision of retinogeniculate inputs (Blitz and Regehr, 2005;Crunelli et al., 1988), the receptive field properties of thalamocortical (TC) neurons, the principal projection neurons of the thalamus (Sillito and Kemp, 1983;Berardi and Morrone, 1984;Holdefer et al., 1989), and dynamically sculpt thalamic network activity (Lorincz et al., 2009). Interneurons innervate TC neurons via conventional axodendritic synapses, or F1 terminals, and dendrodendritic synapses, or F2 terminals, that partake in specialized synaptic "triads" where the interneuron dendrite is postsynaptic to retinogeniculate terminals and presynaptic to the TC neuron dendrite (Famiglietti and Peters, 1972;Grossman et al., 1973;Ohara et al., 1983;Hamos et al., 1985). GABA release from the two terminals differs in its action potential dependence, F1 and F2 terminals being action potential dependent and action potential independent, respectively (Cox and Sherman, 2000;Govindaiah and Cox, 2006), and F2 terminals express both ionotropic and metabotropic glutamate receptors (mGluRs) (Godwin et al., 1996;Cox and Sherman, 2000). Glutamate release from retinogeniculate terminals can therefore monosynaptically excite TC neuron dendrites and disynaptically inhibit them via glutamate receptor-dependent GABA release from F2 terminals (Cox and Sherman, 2000;Govindaiah andCox, 2004, 2006)"

"In the dLGN, enhancement of IPSC frequency and IGABAtonic by group I mGluRs is not action potential dependent, being insensitive to TTX, but is abolished by the L-type Ca2+ channel blocker nimodipine. These results indicate selective mGluR-dependent modulation of dendrodendritic GABA release from F2-type terminals on interneuron dendrites and demonstrate for the first time the presence of eGABAARs on TC neuron dendritic elements that participate in "triadic" circuitry within the dLGN."

(Pressler et al., 2013)

Metabotropic Glutamate Receptors Drive Global Persistent Inhibition in the Visual Thalamus

"Local interneurons in the dLGN regulate the output of TC cells by releasing GABA from their axonal boutons and specialized dendritic spines"

".These local interneurons have long been of interest because, in additional to conventional axodendritic synapses, they inhibit TC cells via specialized dendrodendritic synapses that involve the convergence of an RGC axon, an interneuron dendritic spine, and a relay cell dendrite, collectively known as a triad (Rafols and Valverde, 1973;Ohara et al., 1983;(Ohara et al., 1983). RGC inputs to the interneuron spine are in close proximity to GABA release machinery and to metabotropic glutamate receptors type 5 (mGluR5s) expressed in interneurons (Ohara et al., 1983). This, along with the observation that mGluR5 agonists evoke GABA release from interneuron dendrites in the absence of spiking (Cox et al., 1998;Cox and Sherman, 2000;Govindaiah and Cox, 2004;(Ohara et al., 1983), suggested that RGC axons trigger local GABA release by activating mGluR5 on interneuron spines and locally elevating calcium (Netzeband et al., 1997;Tempia et al., 1998)."

(Govindaiah et al., 2006)

Metabotropic Glutamate Receptors Differentially Regulate GABAergic Inhibition in Thalamus

"Thalamic interneurons and thalamic reticular nucleus (TRN) neurons provide inhibitory innervation of thalamocortical cells that significantly influence thalamic gating. The local interneurons in the dorsal lateral geniculate nucleus (dLGN) give rise to two distinct synaptic outputs: classical axonal and dendrodendritic. Activation of metabotropic glutamate receptors (mGluRs) by agonists or optic tract stimulation increases the output of these presynaptic dendrites leading to increased inhibition of thalamocortical neurons. The present study was aimed to evaluate the actions of specific mGluRs on inhibitory GABA-mediated signaling. We found that the group I mGluR (mGluR1,5) agonist (RS)-3,5-dihydroxyphenylglycine (DHPG) or optic tract stimulation produced a robust increase in spontaneous IPSCs (sIPSCs) in thalamocortical neurons that was attenuated by the selective mGluR5 antagonist 2-methyl-6-(phenylethynyl)pyridine hydrochloride (MPEP). In contrast, the group II mGluR (mGluR2,3) agonists (2R, 4R)-4-aminopyrrolidine-2,4-dicarboxylate (APDC) or (2S,2'R,3'R)-2-(2'3'-dicarboxycyclopropyl)glycine (DCG-IV) suppressed the frequency of sIPSCs. In addition, mGluR1,5 agonist DHPG produced depolarizations and mGluR2/3 agonists APDC or L-CCG-I [(2S,1'S,2'S)-2-(carboxycyclopropyl)glycine] produced hyperpolarizations in dLGN interneurons. Furthermore, the enhanced sIPSC activity by optic tract stimulation was reduced when paired with corticothalamic fiber stimulation. The present data indicate that activation of specific mGluR subtypes differentially regulates inhibitory activity via different synaptic pathways. Our results suggest that activation of specific mGluR subtypes can upregulate or downregulate inhibitory activity in thalamic relay neurons, and these actions likely shape excitatory synaptic integration and thus regulate information transfer through thalamocortical circuits."

(Godwin et al., 1996)

Ultrastructural Localization Suggests that Retinal and Cortical Inputs Access Different Metabotropic Glutamate Receptors in the Lateral Geniculate Nucleus

"Glutamate has an important neuromodulatory role in synaptic transmission through metabotropic glutamate receptors (mGluRs)

linked to a variety of G-protein-coupled second messenger pathways. Activation of these receptors on relay cells in the lateral geniculate nucleus (LGN) with the agonist trans-(1S,3R)−1-amino-1,3-cyclopentanedicarboxylic acid produces a membrane depolarization that inactivates the low-threshold Ca2+ spike, causing a transition from burst to tonic response mode. The excitatory effects of metabotropic receptor activation in the LGN appear to be produced through the receptors linked to phosphoinositide hydrolysis and apparently only through activation of the corticogeniculate pathway. Two mGluRs, mGluR1α (a splice variant of mGluR1) and mGluR5, are linked to the phosphoinositide system. We examined the localization of these receptors with affinity-purified, anti-peptide, polyclonal antibodies raised to the C-terminal region of each receptor protein. Under examination with the light microscope, we found that both types of receptors are present in the geniculate neuropil and in that of the overlying thalamic reticular nucleus, including the perigeniculate nucleus. We also examined the ultrastructural localization of immunolabel with the electron microscope, using a postembedding immunogold marker to identify terminals, dendrites, and somata that contain GABA. Label for the antibody directed against mGluR1α was primarily localized in the dendrites of relay cells, postsynaptic to various terminal types. Of these, terminal profiles normally associated with corticogeniculate inputs predominated, whereas retinal terminal profiles were scarce. Label for the antibody directed against mGluR5 label was prominent in inhibitory F2-terminal profiles associated with the retinal input to relay cells. In the perigeniculate nucleus, both mGluRs were localized to dendrites. The distribution of the two phosphoinositide-linked mGluRs in the LGN suggests very different functional roles for the two receptor types. We conclude from these data that mGluR1 appears to have a dominant role in corticogeniculate control of response mode through the feedback glutamatergic pathway from layer VI, whereas mGluR5 is positioned to affect retinogeniculate activation of relay cells through feed forward glomerular interactions."

(Crandall et al., 2012)

Local dendrodendritic inhibition regulates fast synaptic transmission in visual thalamus

"Inhibition from thalamic interneurons plays a critical role in modulating information transfer between thalamus and neocortex. Interestingly, these neurons yield inhibition via two distinct outputs: presynaptic dendrites that innervate thalamocortical relay neurons and axonal outputs. Since the dendrites of thalamic interneurons are the primary targets of incoming synaptic information, it has been hypothesized that local synaptic input could produce highly focused dendritic output. To gain additional insight into the computational power of these presynaptic dendrites, we have combined two-photon laser scanning microscopy, glutamate uncaging, and whole-cell electrophysiological recordings to locally activate dendritic terminals and study their inhibitory contribution to rat thalamocortical relay neurons. Our findings demonstrate that local dendritic release from thalamic interneurons is controlled locally by AMPA/NMDA receptor-mediated recruitment of L-type calcium channels. Moreover, by mapping these connections with single dendrite resolution we not only found that presynaptic dendrites preferentially target proximal regions, but such actions differ significantly across branches. Furthermore, local stimulation of interneuron dendrites did not result in global excitation, supporting the notion that these interneurons can operate as multiplexors, containing numerous independently operating input–output devices."

(Casale et al., 2011)

Active action potential propagation but not initiation in thalamic interneuron dendrites

"Inhibitory interneurons of the dorsal lateral geniculate nucleus of the thalamus modulate the activity of thalamocortical cells in response to excitatory input through the release of inhibitory neurotransmitter from both axons and dendrites"

".We found that trains or single action potentials elicited by somatic current injection or local synaptic stimulation rapidly and actively backpropagated throughout the entire dendritic arbor and into the fine filiform dendritic appendages known to release GABAergic vesicles"

".Our results indicate that thalamic interneuron dendrites integrate synaptic inputs that initiate action potentials, most likely in the axon initial segment, that then backpropagate with high fidelity into the dendrites, resulting in a nearly synchronous release of GABA from both axonal and dendritic compartments."

(Seabrook et al._1, 2013)

Interneurons in the mouse visual thalamus maintain a high degree of retinal convergence throughout postnatal development

"Far less is known about the pattern of retinal convergence onto interneurons, the other principal cell type present in dLGN. These local-circuit neurons are involved in feedforward inhibition onto relay cells and play a role in contrast gain control, shaping receptive fields of relay cells, and altering the temporal precision of retinal inputs (Sherman, 2004)(Wang et al., 2011)"

".using transgenic mice that express enhanced green fluorescent protein (GFP) in γ-aminobutyric acid (GABA)ergic interneurons (GAD67-GFP) (Chattopadhyaya et al., 2004)"

".GFP-expressing interneurons were evenly distributed throughout dLGN and had highly complex and widespread dendritic processes that often crossed eye-specific borders. Estimates of retinal convergence derived from excitatory postsynaptic potential (EPSP) amplitude by stimulus intensity plots revealed that unlike relay cells, interneurons recorded throughout the first 5 weeks of life, maintain a large number (approximately eight to ten) of retinal inputs. The lack of pruning onto interneurons suggests that the activity-dependent refinement of retinal connections in dLGN is cell-type specific. The high degree of retinal convergence onto interneurons may be necessary for these cells to provide both widespread and local forms of inhibition in dLGN."

(Mullner et al., 2023)

Individual thalamic inhibitory interneurons are functionally specialized toward distinct visual features

> *"The dLGN also contains GABAergic inhibitory interneurons that receive inputs from RGCs and synapse onto TCNs. (Sherman, 2004)"*

> *"Altogether, each inhibitory interneuron globally encodes one visual feature originating mostly in the retina and is ideally suited to perform feature-selective inhibition."*

(Krosigk et al., 1999)

Dynamic properties of corticothalamic excitatory postsynaptic potentials and thalamic reticular inhibitory postsynaptic potentials in thalamocortical neurons of the guinea-pig dorsal lateral geniculate nucleus.

> *"These axons densely innervate thalamocortical neurons in all layers of the LGNd as well as GABAergic neurons, both local and in the thalamic reticular nucleus. (Montero, 2004)(Robson, 1983)(Robson, 1984) revious electrophysiological investigations of the corticothalamic pathway have demonstrated that these axons exhibit a strong monosynaptic excitatory influence on thalamocortical, thalamic reticular, and local GABAergic interneurons. (Ahlsén et al., 1982)(Andersen et al., 1967)(Deschênes et al., 1990)(Kao et al., 1997)(Lindström et al., 2004) The ability of corticothalamic inputs to strongly activate GABAergic neurons within the thalamus also results in the generation of robust di-synaptic inhibitory postsynaptic potentials (IPSPs) in thalamocortical cells, giving the corticothalamic system the ability to either directly excite, or indirectly inhibit, the output of the thalamus."*

(Sanchez-Vives et al., 1996)

Are the Interlaminar Zones of the Ferret Dorsal Lateral Geniculate Nucleus Actually Part of the Perigeniculate Nucleus?

> *"The ferret dorsal lateral geniculate nucleus (LGNd) contains interneurons within the interlaminar zones situated between the laminae corresponding to the ipsi-and contralateral eyes. We found that a subset of these neurons exhibits electrophysiological properties similar to those previously reported for perigeniculate (PGN) neurons, including the generation of rhythmic sequences of rebound low-threshold Ca 2 spikes at a frequency of 1-4 Hz after the intracellular injection of a hyperpolarizing current pulse. These "PGN-like" interlaminar interneurons innervated restricted regions of the A-laminae, inhibited thalamocortical cells through GABA A , and perhaps GABA B , receptors, and were excited by axon collaterals from thalamocortical cells. This reciprocal relationship is identical to that formed by PGN cells and allowed the PGN-like interlaminar neurons to participate in the generation of spindle waves and other network oscillations. Pharmacologically, PGN-like interlaminar interneurons were also similar to PGN neurons: both generated a prolonged depolarization in response to the local application of serotonin, 1S,3R-ACPD, and CCK8S, and a rapid depolarization followed by a more prolonged hyperpolarization in response to acetylcholine"*

> *".Examination of parvalbumin and calbindin staining in the ferret LGNd revealed that both PGN and a subset of interlaminar neurons were parvalbumin-positive. In contrast, calbindinpositive cells were relatively absent in the PGN and sparsely present in the interlaminar zones, but were numerous in the A and C laminae. Our results indicate that the interlaminar zone in between laminae A and A1 and A1 and C in the ferret LGNd possesses a cell type that is electrophysiologically, pharmacologically, anatomically, immunocytochemically, and functionally similar to neurons in the PGN."*

(Whitley et al., 2025)

GABAergic Projections from the Pretectum Boost Retinogeniculate Signal Transfer via Disinhibition

> *"the intrinsic connections of GABAergic dLGN interneurons"*

> *"In the dLGN, synaptic terminals originating from the PT targeted interneurons more frequently than thalamocortical (relay) cells, and optogenetic activation of PT input had a greater impact on interneuron firing frequency compared with relay cells. This cell type-specific impact of PT input to the dLGN resulted in the disinhibition of relay cells and an increase in the amplitude of their postsynaptic responses to retinal input."*

Function and physiological roles

dLGN local interneurons shape what relay cells send to cortex by adding fast feedforward inhibition, highly local triadic inhibition, and, under some conditions, tonic inhibition. Functionally, this lets them control spike count, spike timing, receptive field structure, contrast gain, and parts of thalamic network synchrony. (20 sources)

At the systems level, dLGN local interneurons are best understood as the intrinsic inhibitory cells that tune retinogeniculate transmission rather than merely suppress it. Across classic physiological work and later reviews, they are described as controlling the number of visually evoked spikes in thalamocortical (TC) relay neurons, improving temporal precision of relay firing, sharpening or refining relay-cell receptive fields, and contributing to contrast-gain control of visual responses. These functions follow from their position in the

direct retinal pathway and from the fact that they provide the feedforward inhibitory arm of the dLGN, distinct from recurrent inhibition via perigeniculate/TRN circuits. (Antal et al., 2010) (Seabrook et al., 2013) (Su et al., 2020) (Yang et al., 2017) (Lindstrom et al., 2010) (Halnes et al., 2011)

A key physiological role is fine temporal sculpting of retinal signals. Because retinal excitation to relay dendrites can be paired with near-synchronous disynaptic inhibition through interneuron F2 terminals in triads, local interneurons can narrow the time window in which retinal input drives a relay spike. This is one main reason they are repeatedly linked to improved temporal precision of retinogeniculate signaling. Their dendritic outputs can also act on a branch-by-branch basis, so inhibition can be targeted to single retinal input zones without requiring whole-cell spiking, while backpropagating spikes can recruit more global inhibition when the interneuron is more strongly activated. (Errington et al., 2011) (Govindaiah et al., 2006) (Pressler et al., 2013) (Casale et al., 2011) (Antal et al., 2010) (Yang et al., 2017) (Crandall et al., 2012)

This means the same interneuron can support at least two functional modes. In one mode, local retinal glutamate drives very focal dendrodendritic GABA release at F2 terminals, shaping transmission at individual retinogeniculate contacts and helping refine spatial and temporal filtering. In another mode, spikes initiated near the soma or axon initial segment backpropagate through the dendritic tree and appendages, leading to more coordinated release from both axonal and dendritic compartments and thus broader inhibition across many relay targets. The active properties of the dendrites therefore matter directly for function: postsynaptic excitability in the interneuron helps set the size, spread, and duration of inhibition. (Casale et al., 2011) (Antal et al., 2010) (Pressler et al., 2013)

Metabotropic glutamate signaling gives these cells another layer of control over visual throughput. Group I mGluRs, especially mGluR5 positioned on F2 dendritic terminals in retinal glomeruli, can enhance dendritic GABA release and increase both spontaneous inhibitory events and tonic GABAergic current in relay neurons, in a way that does not require sodium spikes but does require calcium entry through L-type channels. By contrast, group II mGluRs can suppress interneuron-mediated inhibition. Functionally, this means retinal and corticothalamic pathways can push inhibition up or down through receptor-specific mechanisms, allowing state- and pathway-dependent adjustment of excitatory synaptic integration in relay cells. (Errington et al., 2011) (Govindaiah et al., 2006) (Godwin et al., 1996)

Beyond phasic synaptic inhibition, dLGN interneurons can also generate tonic inhibition. Optogenetic work showed that at higher firing rates these cells recruit a strong tonic GABA_A-receptor-mediated conductance in relay neurons via extrasynaptic high-affinity receptors, thereby regulating relay-cell gain. This extends their role from precise point-to-point control at synapses to broader control of excitability across the relay population. (Jager et al., 2016) (Su et al., 2020)

At the network level, local interneurons also help synchronize and pattern thalamic activity. They have been implicated in shaping thalamic oscillations and in rhythmic inhibitory framing of relay-cell output during visually relevant alpha-range activity, where interneuron-mediated phasic inhibition can suppress relay spiking at specific phases of the oscillation. In rodents, recent summaries also describe them as helping spatially and temporally synchronize local dLGN networks. (Halnes et al., 2011) (Errington et al., 2011) (Lorincz et al., 2009) (Weil et al., 2022)

Function may also vary somewhat by species and circuit context. In ferret, a subset of interlaminar “PGN-like” interneurons participates in reciprocal loops with thalamocortical cells and can contribute to spindle-wave and related oscillatory activity, making them functionally closer to perigeniculate neurons than the usual rodent local interneuron description would suggest. In mouse, corticothalamic pathways can recruit interneurons strongly enough to produce disynaptic inhibition of relay cells, and a recent study suggests pyramidal-tract-type cortical input may preferentially drive interneurons and thereby disinhibit relay responses to retinal input through pathway-specific circuit effects. (Sanchez-Vives et al., 1996) (Krosigk et al., 1999) (Whitley et al., 2025)

A useful current synthesis is that each dLGN interneuron may not simply provide generic inhibition, but may

distribute a particular pattern of visually relevant inhibition across many relay cells. Connectomic and modeling-oriented accounts therefore suggest these neurons can combine broad feature-linked influence with highly local triadic computations, making them well suited for feature-selective inhibition within the visual thalamus. (Mullner et al., 2023) (Weil et al., 2022) (Morgan et al., 2019)

Evidence

(Antal et al., 2010)

Cholinergic Activation of M2 Receptors Leads to Context-Dependent Modulation of Feedforward Inhibition in the Visual Thalamus

"In the rodent dLGN, there appears to be just one type of interneuron (Acuña-Goycolea et al., 2008)[16](Williams et al., 1996). The dLGN is a vital link in the visual pathway from the eye to the visual cortex [16,18]. Retinal ganglion cells (RGCs) activate thalamocortical (TC) neurons that convey visually evoked signals to the cortex, and local interneurons within the dLGN that provide feedforward inhibition to TC cells. dLGN interneurons control the number of visually evoked spikes, regulate the precision of firing, and refine the receptive fields (RFs) of TC neurons (Berardi et al., 1984)(Blitz et al., 2005)(Usrey et al., 1999)(Sillito et al., 1983) (Norton et al., 1992)(Norton et al., 1989)"

".dLGN interneurons release GABA from both their axons and dendrites [16](Cox et al., 1998). Their dendrites support both brief sodium action potentials and longlasting calcium spikes, whereas their axons support only sodium spikes. Thus, active responses in the postsynaptic cell control the magnitude, spatial properties and time course of GABA release."

(Seabrook et al., 2013)

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"Far less is known about the pattern of retinal convergence onto interneurons, the other principal cell type present in dLGN. These local-circuit neurons are involved in feedforward inhibition onto relay cells and play a role in contrast gain control, shaping receptive fields of relay cells, and altering the temporal precision of retinal inputs (Sherman, 2004)(Wang et al., 2011)"

".Here we overcame these obstacles by using transgenic mice that express enhanced green fluorescent protein (GFP) in γ-aminobutyric acid (GABA)ergic interneurons (GAD67-GFP) (Chattopadhyaya et al., 2004). Such cell-type specific visualization via GFP allowed us to readily target interneurons for in vitro recordings"

"GFP-expressing interneurons were evenly distributed throughout dLGN and had highly complex and widespread dendritic processes that often crossed eye-specific borders. Estimates of retinal convergence derived from excitatory postsynaptic potential (EPSP) amplitude by stimulus intensity plots revealed that unlike relay cells, interneurons recorded throughout the first 5 weeks of life, maintain a large number (approximately eight to ten) of retinal inputs"

".The lack of pruning onto interneurons suggests that the activity-dependent refinement of retinal connections in dLGN is cell-type specific. The high degree of retinal convergence onto interneurons may be necessary for these cells to provide both widespread and local forms of inhibition in dLGN."

(Su et al., 2020)

Retinal inputs signal astrocytes to recruit interneurons into visual thalamus

"In rodents, the primary retinorecipient regions of visual thalamus include the dorsal lateral geniculate nucleus (dLGN), ventral LGN (vLGN), and intergeniculate leaflet (IGL). These nuclei receive input from a highly diverse group of retinal ganglion cells (RGCs) but only the dLGN serves as a relay of retinal information to the visual cortex (4)(5)(6). Unlike many other regions of rodent thalamus, these three nuclei each contain local GABAergic interneurons (6)(7)(8).

The dLGN, similar to the neocortex, contains a small percentage (10 to 20%) of GABAergic interneurons (8)(9)(10). In contrast to the great diversity of interneuron subtypes in the neocortex (1), dLGN interneurons are homogeneous, clustering into just one or two subtypes based on morphology, intrinsic membrane properties, gene expression, and embryonic origin (11)(12)(13)(14). GABAergic interneurons in dLGN receive monosynaptic input from RGCs and provide feedforward inhibition onto excitatory thalamocortical relay cells (15)(16)(17). Such input serves to sharpen receptive field properties and improve the contrast-gain of retinogeniculate transmission (17)(18)(19)(20)."

(Yang et al., 2017)

Distinct kinetics of inhibitory currents in thalamocortical neurons that arise from dendritic or axonal origin

"Inhibitory inputs to thalamocortical neurons primarily arise from thalamic reticular nucleus (TRN) neurons and local dLGN interneurons. The local interneurons are activated by retinogeniculate afferents, providing feed-forward inhibition to thalamocortical neurons, and these inputs play a significant role in temporal precision and receptive field properties of thalamocortical neurons (Sillito et al., 1983)(Berardi et al., 1984)(Holdefer et al., 1989)(Blitz et al., 2005)"

".The dLGN interneurons are unique and interesting in that they give rise to two distinct types of output onto thalamocortical neurons: classical axonal outputs via F1 terminals and presynaptic dendrites (named F2 terminals) (Guillery, 2004)(Ralston, 1971) (Famiglietti et al., 1972)[18](Cox et al., 1998)(Crandall et al., 2012)(Crandall et al., 2013). Retinal inputs onto thalamocortical neurons provide monosynaptic excitation via a glutamatergic retinogeniculate synapse and disynaptic inhibition via the F2 terminal (presynaptic dendrite of interneuron). These two synapses onto the thalamocortical neurons are within very close proximity, and form a triadic structure [2]22]"

".Local GABAergic interneurons provide two distinct synaptic outputs: axonal (F1 terminals) and dendritic (F2 terminals) onto dLGN thalamocortical neurons. By contrast, TRN neurons provide only axonal output (F1 terminals) onto dLGN thalamocortical neurons."

(Lindstrom et al., 2010)

Feedforward and recurrent inhibitory receptive fields of principal cells in the cat's dorsal lateral geniculate nucleus

"Today, it is known that geniculate principal cells are equipped with two inhibitory circuits: a feedforward inhibitory system with the responsible interneurons located within the main layers of the dLGN and a recurrent inhibitory system with the interneurons confined to the perigeniculate nucleus [18,32]48]. Both types of inhibitory neurons have GABA as their main transmitter. The inhibitory circuits are similar but functionally segregated for X and Y cells (Lindström et al., 1990)."

(Halnes et al., 2011)

A Multi-Compartment Model for Interneurons in the Dorsal Lateral Geniculate Nucleus

"The dorsal lateral geniculate nucleus (dLGN) receives input from retinal ganglion cells and transmits processed information to visual cortex. About 75-80% of the neurons in the dLGN are thalamocortical (TC) neurons, also called relay neurons, as they relay information from the retina to the cortex. Local GABAergic interneurons (INs) constitute the remaining 20-25%, and are responsible for most of the intra-nuclear connections [1]. By providing feed-forward inhibition from retinal ganglion cells to TC neurons, INs control the number of visually evoked spikes in TC neurons, and refine the receptive fields of TC neurons (e.g. (Hamos et al., 1985) (Norton et al., 1992)(Sherman, 2004)(Blitz et al., 2005)). The INs are also important for synchronizing thalamic oscillations (Steriade et al., 1993)(Steriade et al., 1996)"

".For dLGN, a host of physiological and anatomical experiments have provided detailed information about the neuronal connectivity, as well as the morphology and physiology of TC neurons and INs (see e.g. reviews in [22][23][24])"

".We here present a detailed compartmental model of INs using, for the first time, a morphological reconstruction and a set of active dendritic conductances constrained by experimental somatic recordings from INs under several different current-clamp conditions"

".The INs exhibit a rich variety of firing patterns: Depolarizing current injections to the soma may induce tonic firing, periodic bursting or an initial burst followed by tonic spiking, sometimes with prominent spike-time adaptation. When released from hyperpolarization, some INs elicit rebound bursts, while others return more passively to the resting potential."

(Errington et al., 2011)

mGluR control of interneuron output regulates feedforward tonic GABAA inhibition in the visual thalamus

"In the dorsal lateral geniculate nucleus (dLGN), interneurons can alter the temporal precision of retinogeniculate inputs (Blitz and Regehr, 2005;Crunelli et al., 1988), the receptive field properties of thalamocortical (TC) neurons, the principal projection neurons of the thalamus (Sillito and Kemp, 1983;Berardi and Morrone, 1984;Holdefer et al., 1989), and dynamically sculpt thalamic network activity (Lorincz et al., 2009). Interneurons innervate TC neurons via conventional axodendritic synapses, or F1 terminals, and dendrodendritic synapses, or F2 terminals, that partake in specialized synaptic "triads" where the interneuron dendrite is postsynaptic to retinogeniculate terminals and presynaptic to the TC neuron dendrite (Famiglietti and Peters, 1972;Grossman et al., 1973;Ohara et al., 1983;Hamos et al., 1985). GABA release from the two terminals differs in its action potential dependence, F1 and F2 terminals being action potential dependent and action potential independent, respectively (Cox and Sherman, 2000;Govindaiah and Cox, 2006), and F2 terminals express both ionotropic and metabotropic glutamate receptors (mGluRs) (Godwin et al., 1996;Cox and Sherman, 2000). Glutamate release from retinogeniculate terminals can therefore monosynaptically excite TC neuron dendrites and disynaptically inhibit them via glutamate receptor-dependent GABA release from F2 terminals (Cox and Sherman, 2000;Govindaiah andCox, 2004, 2006)"

"In the dLGN, enhancement of IPSC frequency and IGABAtonic by group I mGluRs is not action potential dependent, being insensitive to TTX, but is abolished by the L-type Ca2+ channel blocker nimodipine. These results indicate selective mGluR-dependent modulation of dendrodendritic GABA release from F2-type terminals on interneuron dendrites and demonstrate for the first time the presence of eGABAARs on TC neuron dendritic elements that participate in "triadic" circuitry within the dLGN."

(Govindaiah et al., 2006)

Metabotropic Glutamate Receptors Differentially Regulate GABAergic Inhibition in Thalamus

"Thalamic interneurons and thalamic reticular nucleus (TRN) neurons provide inhibitory innervation of thalamocortical cells that significantly influence thalamic gating. The local interneurons in the dorsal lateral geniculate nucleus (dLGN) give rise to two distinct

synaptic outputs: classical axonal and dendrodendritic. Activation of metabotropic glutamate receptors (mGluRs) by agonists or optic tract stimulation increases the output of these presynaptic dendrites leading to increased inhibition of thalamocortical neurons. The present study was aimed to evaluate the actions of specific mGluRs on inhibitory GABA-mediated signaling. We found that the group I mGluR (mGluR1,5) agonist (RS)-3,5-dihydroxyphenylglycine (DHPG) or optic tract stimulation produced a robust increase in spontaneous IPSCs (sIPSCs) in thalamocortical neurons that was attenuated by the selective mGluR5 antagonist 2-methyl-6-(phenylethynyl)pyridine hydrochloride (MPEP). In contrast, the group II mGluR (mGluR2,3) agonists (2R, 4R)-4-aminopyrrolidine-2,4-dicarboxylate (APDC) or (2S,2'R,3'R)-2-(2'3'-dicarboxycyclopropyl)glycine (DCG-IV) suppressed the frequency of sIPSCs. In addition, mGluR1,5 agonist DHPG produced depolarizations and mGluR2/3 agonists APDC or L-CCG-I [(2S,1'S,2'S)-2-(carboxycyclopropyl)glycine] produced hyperpolarizations in dLGN interneurons. Furthermore, the enhanced sIPSC activity by optic tract stimulation was reduced when paired with corticothalamic fiber stimulation. The present data indicate that activation of specific mGluR subtypes differentially regulates inhibitory activity via different synaptic pathways. Our results suggest that activation of specific mGluR subtypes can upregulate or downregulate inhibitory activity in thalamic relay neurons, and these actions likely shape excitatory synaptic integration and thus regulate information transfer through thalamocortical circuits."

(Pressler et al., 2013)

Metabotropic Glutamate Receptors Drive Global Persistent Inhibition in the Visual Thalamus

"Local interneurons in the dLGN regulate the output of TC cells by releasing GABA from their axonal boutons and specialized dendritic spines"

".These local interneurons have long been of interest because, in additional to conventional axodendritic synapses, they inhibit TC cells via specialized dendrodendritic synapses that involve the convergence of an RGC axon, an interneuron dendritic spine, and a relay cell dendrite, collectively known as a triad (Rafols and Valverde, 1973;Ohara et al., 1983;(Ohara et al., 1983). RGC inputs to the interneuron spine are in close proximity to GABA release machinery and to metabotropic glutamate receptors type 5 (mGluR5s) expressed in interneurons (Ohara et al., 1983). This, along with the observation that mGluR5 agonists evoke GABA release from interneuron dendrites in the absence of spiking (Cox et al., 1998;Cox and Sherman, 2000;Govindaiah and Cox, 2004;(Ohara et al., 1983), suggested that RGC axons trigger local GABA release by activating mGluR5 on interneuron spines and locally elevating calcium (Netzeband et al., 1997;Tempia et al., 1998)."

(Casale et al., 2011)

Active action potential propagation but not initiation in thalamic interneuron dendrites

"Inhibitory interneurons of the dorsal lateral geniculate nucleus of the thalamus modulate the activity of thalamocortical cells in response to excitatory input through the release of inhibitory neurotransmitter from both axons and dendrites"

".We found that trains or single action potentials elicited by somatic current injection or local synaptic stimulation rapidly and actively backpropagated throughout the entire dendritic arbor and into the fine filiform dendritic appendages known to release GABAergic vesicles"

".Our results indicate that thalamic interneuron dendrites integrate synaptic inputs that initiate action potentials, most likely in the axon initial segment, that then backpropagate with high fidelity into the dendrites, resulting in a nearly synchronous release of GABA from both axonal and dendritic compartments."

(Crandall et al., 2012)

Local dendrodendritic inhibition regulates fast synaptic transmission in visual thalamus

"Inhibition from thalamic interneurons plays a critical role in modulating information transfer between thalamus and neocortex. Interestingly, these neurons yield inhibition via two distinct outputs: presynaptic dendrites that innervate thalamocortical relay neurons and axonal outputs. Since the dendrites of thalamic interneurons are the primary targets of incoming synaptic information, it has been hypothesized that local synaptic input could produce highly focused dendritic output. To gain additional insight into the computational power of these presynaptic dendrites, we have combined two-photon laser scanning microscopy, glutamate uncaging, and whole-cell electrophysiological recordings to locally activate dendritic terminals and study their inhibitory contribution to rat thalamocortical relay neurons. Our findings demonstrate that local dendritic release from thalamic interneurons is controlled locally by AMPA/NMDA receptor-mediated recruitment of L-type calcium channels. Moreover, by mapping these connections with single dendrite resolution we not only found that presynaptic dendrites preferentially target proximal regions, but such actions differ significantly across branches. Furthermore, local stimulation of interneuron dendrites did not result in global excitation, supporting the notion that these interneurons can operate as multiplexors, containing numerous independently operating input–output devices."

(Godwin et al., 1996)

Ultrastructural Localization Suggests that Retinal and Cortical Inputs Access Different Metabotropic Glutamate Receptors in the Lateral Geniculate Nucleus

"Glutamate has an important neuromodulatory role in synaptic transmission through metabotropic glutamate receptors (mGluRs) linked to a variety of G-protein-coupled second messenger pathways. Activation of these receptors on relay cells in the lateral geniculate nucleus (LGN) with the agonist trans-(1S,3R)−1-amino-1,3-cyclopentanedicarboxylic acid produces a membrane

depolarization that inactivates the low-threshold Ca2+ spike, causing a transition from burst to tonic response mode. The excitatory effects of metabotropic receptor activation in the LGN appear to be produced through the receptors linked to phosphoinositide hydrolysis and apparently only through activation of the corticogeniculate pathway. Two mGluRs, mGluR1α (a splice variant of mGluR1) and mGluR5, are linked to the phosphoinositide system. We examined the localization of these receptors with affinity-purified, anti-peptide, polyclonal antibodies raised to the C-terminal region of each receptor protein. Under examination with the light microscope, we found that both types of receptors are present in the geniculate neuropil and in that of the overlying thalamic reticular nucleus, including the perigeniculate nucleus. We also examined the ultrastructural localization of immunolabel with the electron microscope, using a postembedding immunogold marker to identify terminals, dendrites, and somata that contain GABA. Label for the antibody directed against mGluR1α was primarily localized in the dendrites of relay cells, postsynaptic to various terminal types. Of these, terminal profiles normally associated with corticogeniculate inputs predominated, whereas retinal terminal profiles were scarce. Label for the antibody directed against mGluR5 label was prominent in inhibitory F2-terminal profiles associated with the retinal input to relay cells. In the perigeniculate nucleus, both mGluRs were localized to dendrites. The distribution of the two phosphoinositide-linked mGluRs in the LGN suggests very different functional roles for the two receptor types. We conclude from these data that mGluR1 appears to have a dominant role in corticogeniculate control of response mode through the feedback glutamatergic pathway from layer VI, whereas mGluR5 is positioned to affect retinogeniculate activation of relay cells through feed forward glomerular interactions."

(Jager et al., 2016)

Tectal-derived interneurons contribute to phasic and tonic inhibition in the visual thalamus

"The release of GABA from local interneurons in the dorsal lateral geniculate nucleus (dLGN-INs) provides inhibitory control during visual processing within the thalamus"

". during early postnatal development Sox14/Otx2-expressing precursor cells migrate from the dorsal midbrain to generate dLGN-INs. The unexpected extra-diencephalic origin of dLGN-INs sets them apart from GABAergic neurons of the reticular thalamic nucleus. Using optogenetics we show that at increased firing rates tectal-derived dLGN-INs generate a powerful form of tonic inhibition that regulates the gain of thalamic relay neurons through recruitment of extrasynaptic high-affinity GABAA receptors"

". Although interneurons are ubiquitous in the thalamus of carnivorans and primates, they are largely confined to the dorsal lateral geniculate nucleus (dLGN) in rodents 28 . Retinal input to the dLGN directly excites local GABAergic interneurons (dLGN-INs) [29][30] [31] providing visually driven feed-forward inhibition of dLGN thalamic relay neurons 32 . The F2 terminals formed by dLGN-INs also display atypical wiring, releasing GABA at dendro-dendritic sites 33 found within triadic structures known as glomeruli 34 . In addition, F1 terminals are more classical axo-dendrtic synaptic arrangements."

(Lorincz et al., 2009)

Temporal Framing of Thalamic Relay-Mode Firing by Phasic Inhibition during the Alpha Rhythm

"Summary Several aspects of perception, particularly those pertaining to vision, are closely linked to the occipital alpha (α) rhythm. However, how the α rhythm relates to the activity of neurons that convey primary visual information is unknown. Here we show that in behaving cats, thalamocortical neurons in the lateral geniculate nucleus (LGN) that operate in a conventional relay-mode form two groups where the cumulative firing is subject to a cyclic suppression that is centered on the negative α rhythm peak in one group and on the positive peak in the other. This leads to an effective temporal framing of relay-mode output and results from phasic inhibition from LGN interneurons, which in turn are rhythmically excited by thalamocortical neurons that exhibit high-threshold bursts. These results provide a potential cellular substrate for linking the α rhythm to perception and further underscore the central role of inhibition in controlling spike timing during cognitively relevant brain oscillations."

(Weil et al., 2022)

Daily changes in light influence mood via inhibitory networks within the thalamic perihabenular nucleus

"Unlike the majority of the dorsal thalamus in mice, the retino-recipient dorsal lateral geniculate nucleus (dLGN) contains a subset of inhibitory (GABAergic) interneurons that spatially and temporally synchronize local networks (15)(Halassa et al., 2019)(Sherman, 2004)"

".These long-range inhibitory connections differentiate the PHb from the dLGN, which only contains local inhibitory networks (Morgan et al., 2019)."

(Sanchez-Vives et al., 1996)

Are the Interlaminar Zones of the Ferret Dorsal Lateral Geniculate Nucleus Actually Part of the Perigeniculate Nucleus?

"The ferret dorsal lateral geniculate nucleus (LGNd) contains interneurons within the interlaminar zones situated between the laminae corresponding to the ipsi-and contralateral eyes. We found that a subset of these neurons exhibits electrophysiological properties similar to those previously reported for perigeniculate (PGN) neurons, including the generation of rhythmic sequences of rebound low-threshold Ca 2 spikes at a frequency of 1-4 Hz after the intracellular injection of a hyperpolarizing current pulse. These "PGN-like" interlaminar interneurons innervated restricted regions of the A-laminae, inhibited thalamocortical cells through GABA A , and perhaps GABA B , receptors, and were excited by axon collaterals from thalamocortical cells. This reciprocal relationship is

identical to that formed by PGN cells and allowed the PGN-like interlaminar neurons to participate in the generation of spindle waves and other network oscillations. Pharmacologically, PGN-like interlaminar interneurons were also similar to PGN neurons: both generated a prolonged depolarization in response to the local application of serotonin, 1S,3R-ACPD, and CCK8S, and a rapid depolarization followed by a more prolonged hyperpolarization in response to acetylcholine"

".Examination of parvalbumin and calbindin staining in the ferret LGNd revealed that both PGN and a subset of interlaminar neurons were parvalbumin-positive. In contrast, calbindinpositive cells were relatively absent in the PGN and sparsely present in the interlaminar zones, but were numerous in the A and C laminae. Our results indicate that the interlaminar zone in between laminae A and A1 and A1 and C in the ferret LGNd possesses a cell type that is electrophysiologically, pharmacologically, anatomically, immunocytochemically, and functionally similar to neurons in the PGN."

(Krosigk et al., 1999)

Dynamic properties of corticothalamic excitatory postsynaptic potentials and thalamic reticular inhibitory postsynaptic potentials in thalamocortical neurons of the guinea-pig dorsal lateral geniculate nucleus.

"These axons densely innervate thalamocortical neurons in all layers of the LGNd as well as GABAergic neurons, both local and in the thalamic reticular nucleus. (Montero, 2004)(Robson, 1983)(Robson, 1984) revious electrophysiological investigations of the corticothalamic pathway have demonstrated that these axons exhibit a strong monosynaptic excitatory influence on thalamocortical, thalamic reticular, and local GABAergic interneurons. (Ahlsén et al., 1982)(Andersen et al., 1967)(Deschênes et al., 1990)(Kao et al., 1997)(Lindström et al., 2004) The ability of corticothalamic inputs to strongly activate GABAergic neurons within the thalamus also results in the generation of robust di-synaptic inhibitory postsynaptic potentials (IPSPs) in thalamocortical cells, giving the corticothalamic system the ability to either directly excite, or indirectly inhibit, the output of the thalamus."

(Whitley et al., 2025)

GABAergic Projections from the Pretectum Boost Retinogeniculate Signal Transfer via Disinhibition

"the intrinsic connections of GABAergic dLGN interneurons"

"In the dLGN, synaptic terminals originating from the PT targeted interneurons more frequently than thalamocortical (relay) cells, and optogenetic activation of PT input had a greater impact on interneuron firing frequency compared with relay cells. This cell type-specific impact of PT input to the dLGN resulted in the disinhibition of relay cells and an increase in the amplitude of their postsynaptic responses to retinal input."

(Mullner et al., 2023)

Individual thalamic inhibitory interneurons are functionally specialized toward distinct visual features

"The dLGN also contains GABAergic inhibitory interneurons that receive inputs from RGCs and synapse onto TCNs. (Sherman, 2004)"

"Altogether, each inhibitory interneuron globally encodes one visual feature originating mostly in the retina and is ideally suited to perform feature-selective inhibition."

(Morgan et al., 2019)

An individual interneuron participates in many kinds of inhibition and innervates much of the mouse visual thalamus

"One way to assess a neuron's function is to describe all its inputs and outputs. With this goal in mind, we used serial section electron microscopy to map 899 synaptic inputs and 623 outputs in one inhibitory interneuron in a large volume of the mouse visual thalamus. This neuron innervated 256 thalamocortical cells spread across functionally distinct subregions of the visual thalamus. All but one of its neurites were bifunctional, innervating thalamocortical and local interneurons while also receiving synapses from the retina. We observed a wide variety of local synaptic motifs. While this neuron innervated many cells weakly, with single en passant synapses, it also deployed specialized branches that climbed along other dendrites to form strong multi-synaptic connections with a subset of partners. This neuron's diverse range of synaptic relationships allows it to participate in a mix of global and local processing but defies assigning it a single circuit function."

Neurotransmitters, neuropeptides, and synaptic mechanisms

dLGN local interneurons are GABAergic cells that inhibit relay neurons through both axonal boutons and presynaptic dendrites. Their dendritic output is unusual because it can be triggered locally by retinal glutamate and mGluR5-dependent calcium signaling without a full somatic spike, while stronger activation can synchronize release from both dendrites and axons. (13 sources)

The main fast transmitter used by dLGN local interneurons is GABA, and both the local feedforward inhibitory system inside dLGN and the recurrent perigeniculate/TRN inhibitory system are GABAergic. In dLGN

ultrastructure, their inhibitory boutons belong to the class of F terminals with flattened vesicles, and local interneurons are one of the two sources of these GABAergic terminals in the nucleus. (Lindstrom et al., 2010) (Hammer et al., 2014)

What sets dLGN interneurons apart is not a different transmitter, but where and how they release it. They release GABA from conventional axonal boutons as well as from specialized presynaptic dendrites, so a single cell has both F1 axodendritic outputs and F2 dendrodendritic outputs onto thalamocortical relay neurons. F2 terminals sit in retinal glomeruli/triads, where an RGC terminal excites both the relay dendrite and the interneuron dendritic appendage, allowing retinal glutamate to recruit closely matched disynaptic inhibition. (Errington et al., 2011) (Pressler et al., 2013) (Yang et al., 2017) (Jager et al., 2016)

These two output compartments follow different release rules. F1 output is action-potential dependent, but F2 dendritic output can be action-potential independent and driven locally by glutamatergic input onto the presynaptic dendrite itself. In these dendrites, retinal glutamate can act through ionotropic glutamate receptors and through group I metabotropic glutamate receptors, especially mGluR5, to raise local calcium and trigger GABA release. (Errington et al., 2011) (Govindaiah et al., 2006) (Godwin et al., 1996) (Crandall et al., 2012) (Pressler et al., 2013)

The receptor logic of this dendritic output has been studied in some detail. mGluR5 protein is enriched in inhibitory F2 profiles associated with retinal input, placing it in the right subcellular site to control feedforward dendritic GABA release, whereas mGluR1α is found mainly in relay-cell dendrites and is linked more to corticogeniculate control. Pharmacologically, activating group I mGluRs with DHPG or by optic tract stimulation increases spontaneous IPSCs in relay neurons, and this effect is reduced by an mGluR5 antagonist. By contrast, group II mGluRs suppress interneuron-mediated inhibitory activity and hyperpolarize interneurons, while group I agonists depolarize them. (Godwin et al., 1996) (Govindaiah et al., 2006) (Errington et al., 2011)

Calcium entry is a key step in this mechanism. Group I mGluR-driven enhancement of IPSC frequency and tonic GABA current in relay neurons is insensitive to TTX but blocked by the L-type calcium channel blocker nimodipine, supporting a local dendritic release mechanism rather than one that depends on propagated sodium spikes. At single-branch scale, local glutamatergic stimulation of interneuron dendrites recruits AMPA/NMDA receptors and L-type calcium channels and can produce branch-specific inhibitory output without global cell excitation, consistent with the idea that different dendritic branches can function as semi-independent input-output units. (Errington et al., 2011) (Crandall et al., 2012)

Still, these cells are not limited to purely local dendritic signaling. Action potentials initiated by somatic current injection or synaptic stimulation can backpropagate rapidly through the whole dendritic arbor and into the fine GABA-releasing appendages, so stronger activation can produce nearly synchronous GABA release from both axonal and dendritic compartments. This matches the broader view that active dendrites help set the magnitude, spread, and time course of inhibition. (Casale et al., 2011) (Antal et al., 2010)

The inhibitory effects seen by relay neurons include both fast synaptic inhibition and a tonic component. During development, optic tract stimulation evokes an EPSP/IPSP sequence in which the IPSP has both GABA_A and GABA_B components by around P14, alongside robust GABA immunostaining and mature synaptic profiles. In adults, higher firing rates in tectum-derived dLGN interneurons can recruit a strong tonic inhibition through extrasynaptic high-affinity GABA_A receptors on relay neurons, adding a gain-control mode on top of point-to-point synaptic inhibition. (Bickford et al., 2010) (Hammer et al., 2014) (Jager et al., 2016)

As for neuropeptides, the provided dLGN literature does not identify a canonical peptide transmitter for the main rodent local interneuron class. Instead, peptide-related data in these sources are limited to modulatory sensitivity in some species: in ferret, a PGN-like subset of interlaminar interneurons responds to CCK8S as well as serotonin and acetylcholine, and inhibits thalamocortical cells through GABA_A, and perhaps GABA_B, receptors. This looks like a subtype- and species-specific pharmacological feature rather than an established peptide identity for rodent dLGN local interneurons as a whole. (Sanchez-Vives et al., 1996)

Evidence

(Lindstrom et al., 2010)

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(Hammer et al., 2014)

Nuclei-specific differences in nerve terminal distribution, morphology, and development in mouse visual thalamus

"The overwhelming majority of these studies have focused on inputs to and projections from the dorsal lateral geniculate nucleus (dLGN). Relay neurons within dLGN receive strong glutamatergic inputs from retinal ganglion cells (RGCs) and serve as the principle conduit of visual signaling to the cortex. However, relay neurons do not act as passive relays of visual information. The gain of retinogeniculate signal transmission is modulated by nonretinal inputs to dLGN. These nonretinal inputs arise from visual cortex, pretectum, brainstem, thalamic reticular nuclei, and local dLGN interneurons, and they far outnumber the more powerful retinal inputs (Sherman et al., 2002)(Sherman, 2004). In fact, nonretinal inputs account for as much as 95% of the nerve terminals in dLGN (Sherman et al., 2002)(Erisir et al., 1997)(Erisir et al., 1997)(Horn et al., 2000)(Bickford et al., 2010)"

".One set of nerve terminals contain flattened, oval shaped synaptic vesicles, a hallmark feature of inhibitory GABAergic terminals, and are therefore called F terminals (Sherman, 2004)(Lund et al., 1972)(Famiglietti, 1970). In dLGN, GABAergic terminals arise from the thalamic reticular nucleus and local inhibitory interneurons [7,18]19]."

(Errington et al., 2011)

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(Yang et al., 2017)

Distinct kinetics of inhibitory currents in thalamocortical neurons that arise from dendritic or axonal origin

"Inhibitory inputs to thalamocortical neurons primarily arise from thalamic reticular nucleus (TRN) neurons and local dLGN interneurons. The local interneurons are activated by retinogeniculate afferents, providing feed-forward inhibition to thalamocortical neurons, and these inputs play a significant role in temporal precision and receptive field properties of thalamocortical neurons (Sillito et al., 1983)(Berardi et al., 1984)(Holdefer et al., 1989)(Blitz et al., 2005)"

".The dLGN interneurons are unique and interesting in that they give rise to two distinct types of output onto thalamocortical neurons: classical axonal outputs via F1 terminals and presynaptic dendrites (named F2 terminals) (Guillery, 2004)(Ralston, 1971) (Famiglietti et al., 1972)[18](Cox et al., 1998)(Crandall et al., 2012)(Crandall et al., 2013). Retinal inputs onto thalamocortical neurons provide monosynaptic excitation via a glutamatergic retinogeniculate synapse and disynaptic inhibition via the F2 terminal (presynaptic dendrite of interneuron). These two synapses onto the thalamocortical neurons are within very close proximity, and form a triadic structure [2]22]"

".Local GABAergic interneurons provide two distinct synaptic outputs: axonal (F1 terminals) and dendritic (F2 terminals) onto dLGN thalamocortical neurons. By contrast, TRN neurons provide only axonal output (F1 terminals) onto dLGN thalamocortical neurons."

(Jager et al., 2016)

Tectal-derived interneurons contribute to phasic and tonic inhibition in the visual thalamus

"The release of GABA from local interneurons in the dorsal lateral geniculate nucleus (dLGN-INs) provides inhibitory control during visual processing within the thalamus"

". during early postnatal development Sox14/Otx2-expressing precursor cells migrate from the dorsal midbrain to generate dLGN-INs. The unexpected extra-diencephalic origin of dLGN-INs sets them apart from GABAergic neurons of the reticular thalamic nucleus. Using optogenetics we show that at increased firing rates tectal-derived dLGN-INs generate a powerful form of tonic inhibition that regulates the gain of thalamic relay neurons through recruitment of extrasynaptic high-affinity GABAA receptors"

". Although interneurons are ubiquitous in the thalamus of carnivorans and primates, they are largely confined to the dorsal lateral geniculate nucleus (dLGN) in rodents 28 . Retinal input to the dLGN directly excites local GABAergic interneurons (dLGN-INs) [29][30] [31] providing visually driven feed-forward inhibition of dLGN thalamic relay neurons 32 . The F2 terminals formed by dLGN-INs also display atypical wiring, releasing GABA at dendro-dendritic sites 33 found within triadic structures known as glomeruli 34 . In addition, F1 terminals are more classical axo-dendrtic synaptic arrangements."

(Govindaiah et al., 2006)

Metabotropic Glutamate Receptors Differentially Regulate GABAergic Inhibition in Thalamus

"Thalamic interneurons and thalamic reticular nucleus (TRN) neurons provide inhibitory innervation of thalamocortical cells that significantly influence thalamic gating. The local interneurons in the dorsal lateral geniculate nucleus (dLGN) give rise to two distinct synaptic outputs: classical axonal and dendrodendritic. Activation of metabotropic glutamate receptors (mGluRs) by agonists or optic tract stimulation increases the output of these presynaptic dendrites leading to increased inhibition of thalamocortical neurons. The present study was aimed to evaluate the actions of specific mGluRs on inhibitory GABA-mediated signaling. We found that the group I mGluR (mGluR1,5) agonist (RS)-3,5-dihydroxyphenylglycine (DHPG) or optic tract stimulation produced a robust increase in spontaneous IPSCs (sIPSCs) in thalamocortical neurons that was attenuated by the selective mGluR5 antagonist 2-methyl-6-(phenylethynyl)pyridine hydrochloride (MPEP). In contrast, the group II mGluR (mGluR2,3) agonists (2R, 4R)-4-aminopyrrolidine-2,4-dicarboxylate (APDC) or (2S,2'R,3'R)-2-(2'3'-dicarboxycyclopropyl)glycine (DCG-IV) suppressed the frequency of sIPSCs. In addition, mGluR1,5 agonist DHPG produced depolarizations and mGluR2/3 agonists APDC or L-CCG-I [(2S,1'S,2'S)-2-(carboxycyclopropyl)glycine] produced hyperpolarizations in dLGN interneurons. Furthermore, the enhanced sIPSC activity by optic tract stimulation was reduced when paired with corticothalamic fiber stimulation. The present data indicate that activation of specific mGluR subtypes differentially regulates inhibitory activity via different synaptic pathways. Our results suggest that activation of specific mGluR subtypes can upregulate or downregulate inhibitory activity in thalamic relay neurons, and these actions likely shape excitatory synaptic integration and thus regulate information transfer through thalamocortical circuits."

(Godwin et al., 1996)

Ultrastructural Localization Suggests that Retinal and Cortical Inputs Access Different Metabotropic Glutamate Receptors in the Lateral Geniculate Nucleus

"Glutamate has an important neuromodulatory role in synaptic transmission through metabotropic glutamate receptors (mGluRs) linked to a variety of G-protein-coupled second messenger pathways. Activation of these receptors on relay cells in the lateral geniculate nucleus (LGN) with the agonist trans-(1S,3R)−1-amino-1,3-cyclopentanedicarboxylic acid produces a membrane depolarization that inactivates the low-threshold Ca2+ spike, causing a transition from burst to tonic response mode. The excitatory effects of metabotropic receptor activation in the LGN appear to be produced through the receptors linked to phosphoinositide hydrolysis and apparently only through activation of the corticogeniculate pathway. Two mGluRs, mGluR1α (a splice variant of mGluR1) and mGluR5, are linked to the phosphoinositide system. We examined the localization of these receptors with affinity-purified, anti-peptide, polyclonal antibodies raised to the C-terminal region of each receptor protein. Under examination with the light microscope, we found that both types of receptors are present in the geniculate neuropil and in that of the overlying thalamic reticular nucleus, including the perigeniculate nucleus. We also examined the ultrastructural localization of immunolabel with the

electron microscope, using a postembedding immunogold marker to identify terminals, dendrites, and somata that contain GABA. Label for the antibody directed against mGluR1α was primarily localized in the dendrites of relay cells, postsynaptic to various terminal types. Of these, terminal profiles normally associated with corticogeniculate inputs predominated, whereas retinal terminal profiles were scarce. Label for the antibody directed against mGluR5 label was prominent in inhibitory F2-terminal profiles associated with the retinal input to relay cells. In the perigeniculate nucleus, both mGluRs were localized to dendrites. The distribution of the two phosphoinositide-linked mGluRs in the LGN suggests very different functional roles for the two receptor types. We conclude from these data that mGluR1 appears to have a dominant role in corticogeniculate control of response mode through the feedback glutamatergic pathway from layer VI, whereas mGluR5 is positioned to affect retinogeniculate activation of relay cells through feed forward glomerular interactions."

(Crandall et al., 2012)

Local dendrodendritic inhibition regulates fast synaptic transmission in visual thalamus

"Inhibition from thalamic interneurons plays a critical role in modulating information transfer between thalamus and neocortex. Interestingly, these neurons yield inhibition via two distinct outputs: presynaptic dendrites that innervate thalamocortical relay neurons and axonal outputs. Since the dendrites of thalamic interneurons are the primary targets of incoming synaptic information, it has been hypothesized that local synaptic input could produce highly focused dendritic output. To gain additional insight into the computational power of these presynaptic dendrites, we have combined two-photon laser scanning microscopy, glutamate uncaging, and whole-cell electrophysiological recordings to locally activate dendritic terminals and study their inhibitory contribution to rat thalamocortical relay neurons. Our findings demonstrate that local dendritic release from thalamic interneurons is controlled locally by AMPA/NMDA receptor-mediated recruitment of L-type calcium channels. Moreover, by mapping these connections with single dendrite resolution we not only found that presynaptic dendrites preferentially target proximal regions, but such actions differ significantly across branches. Furthermore, local stimulation of interneuron dendrites did not result in global excitation, supporting the notion that these interneurons can operate as multiplexors, containing numerous independently operating input–output devices."

(Casale et al., 2011)

Active action potential propagation but not initiation in thalamic interneuron dendrites

"Inhibitory interneurons of the dorsal lateral geniculate nucleus of the thalamus modulate the activity of thalamocortical cells in response to excitatory input through the release of inhibitory neurotransmitter from both axons and dendrites"

".We found that trains or single action potentials elicited by somatic current injection or local synaptic stimulation rapidly and actively backpropagated throughout the entire dendritic arbor and into the fine filiform dendritic appendages known to release GABAergic vesicles"

".Our results indicate that thalamic interneuron dendrites integrate synaptic inputs that initiate action potentials, most likely in the axon initial segment, that then backpropagate with high fidelity into the dendrites, resulting in a nearly synchronous release of GABA from both axonal and dendritic compartments."

(Antal et al., 2010)

Cholinergic Activation of M2 Receptors Leads to Context-Dependent Modulation of Feedforward Inhibition in the Visual Thalamus

"In the rodent dLGN, there appears to be just one type of interneuron (Acuña-Goycolea et al., 2008)[16](Williams et al., 1996). The dLGN is a vital link in the visual pathway from the eye to the visual cortex [16,18]. Retinal ganglion cells (RGCs) activate thalamocortical (TC) neurons that convey visually evoked signals to the cortex, and local interneurons within the dLGN that provide feedforward inhibition to TC cells. dLGN interneurons control the number of visually evoked spikes, regulate the precision of firing, and refine the receptive fields (RFs) of TC neurons (Berardi et al., 1984)(Blitz et al., 2005)(Usrey et al., 1999)(Sillito et al., 1983) (Norton et al., 1992)(Norton et al., 1989)"

".dLGN interneurons release GABA from both their axons and dendrites [16](Cox et al., 1998). Their dendrites support both brief sodium action potentials and longlasting calcium spikes, whereas their axons support only sodium spikes. Thus, active responses in the postsynaptic cell control the magnitude, spatial properties and time course of GABA release."

(Bickford et al., 2010)

Synaptic Development of the Mouse Dorsal Lateral Geniculate Nucleus

"The dorsal lateral geniculate nucleus (dLGN) of the mouse has emerged as a model system in the study of thalamic circuit development. However, there is still a lack of information regarding how and when various types of retinal and nonretinal synapses develop. We examined the synaptic organization of the developing mouse dLGN in the common pigmented C57/BL6 strain, by recording the synaptic responses evoked by electrical stimulation of optic tract axons, and by investigating the ultrastructure of identified synapses. At early postnatal ages (P14), when optic tract stimulation routinely evoked an excitatory postsynaptic potential/inhibitory postsynaptic potential (EPSP/IPSP) sequence, with the latter having both a GABAA and GABAB component. Electrophysiological and ultrastructural observations were consistent. At P7, many synapses were present, but synaptic profiles lacked the ultrastructural features characteristic of the adult dLGN, and little γ-aminobutyric acid (GABA) could be detected by using immunocytochemical techniques. In contrast, by P14, GABA staining was robust, mature synaptic profiles of retinal and nonretinal

origin were easily distinguished, and the size and proportion of synaptic contacts were similar to those of the adult. The emergence of nonretinal synapses coincides with pruning of retinogeniculate connections, and the transition of retinal activity from spontaneous to visually driven. These results indicate that the synaptic architecture of the mouse dLGN is similar to that of other higher mammals, and thus provides further support for its use as a model system for visual system development. J. Comp. Neurol. 518:622–635, 2010. © 2009 Wiley-Liss, Inc."

(Sanchez-Vives et al., 1996)

Are the Interlaminar Zones of the Ferret Dorsal Lateral Geniculate Nucleus Actually Part of the Perigeniculate Nucleus?

"The ferret dorsal lateral geniculate nucleus (LGNd) contains interneurons within the interlaminar zones situated between the laminae corresponding to the ipsi-and contralateral eyes. We found that a subset of these neurons exhibits electrophysiological properties similar to those previously reported for perigeniculate (PGN) neurons, including the generation of rhythmic sequences of rebound low-threshold Ca 2 spikes at a frequency of 1-4 Hz after the intracellular injection of a hyperpolarizing current pulse. These "PGN-like" interlaminar interneurons innervated restricted regions of the A-laminae, inhibited thalamocortical cells through GABA A , and perhaps GABA B , receptors, and were excited by axon collaterals from thalamocortical cells. This reciprocal relationship is identical to that formed by PGN cells and allowed the PGN-like interlaminar neurons to participate in the generation of spindle waves and other network oscillations. Pharmacologically, PGN-like interlaminar interneurons were also similar to PGN neurons: both generated a prolonged depolarization in response to the local application of serotonin, 1S,3R-ACPD, and CCK8S, and a rapid depolarization followed by a more prolonged hyperpolarization in response to acetylcholine"

".Examination of parvalbumin and calbindin staining in the ferret LGNd revealed that both PGN and a subset of interlaminar neurons were parvalbumin-positive. In contrast, calbindinpositive cells were relatively absent in the PGN and sparsely present in the interlaminar zones, but were numerous in the A and C laminae. Our results indicate that the interlaminar zone in between laminae A and A1 and A1 and C in the ferret LGNd possesses a cell type that is electrophysiologically, pharmacologically, anatomically, immunocytochemically, and functionally similar to neurons in the PGN."

Markers and molecular identity

There is no long list of firmly established subtype markers for the main rodent dLGN local interneuron class in the material provided. The most useful markers here are GABA/GAD-based labels for identifying the class, developmental markers such as Sox14 and Otx2 for lineage, and a few context-specific markers such as mGluR5 at F2 dendritic terminals and parvalbumin in a ferret PGN-like subset. (8 sources)

- **GABA / GABA-immunoreactivity**
- **Type:** protein / transmitter immunolabel.
- **Context:** class marker for identifying inhibitory local interneurons within mouse dLGN in histological counts.
- **Use:** dLGN interneurons were identified as the main intrinsic GABAergic population of mouse dorsal thalamus, and stereological work used **GABA immunolabeling** to identify these cells when estimating their abundance. This is a marker for the inhibitory class, not for a known subtype within rodent dLGN interneurons. (Somaiya et al., 2022) (Evangelio et al., 2018)
- **GAD67-GFP**
- **Type:** transgene reporter.
- **Context:** experimental marker used in mouse to visualize and target dLGN interneurons for recording and developmental morphology.
- **Use:** multiple studies used **GAD67-GFP mice** to label dLGN GABAergic interneurons, allowing targeted recordings and reconstruction. In this context, GFP is not itself a native molecular identity marker; it is a transgenic readout driven by the GAD67/Gad1 GABAergic program and serves as a practical label for the interneuron class in mouse dLGN. (Seabrook et al., 2013) (Chattopadhyaya et al., 2004) (Charalambakis et al., 2019)
- **GAD67 / Gad1**
- **Type:** protein/transcript basis inferred from the GAD67-GFP line.

- **Context:** broad GABAergic-class marker rather than a dLGN-IN subtype marker.
- **Use:** the use of the GAD67-GFP line implies that dLGN local interneurons express the GABA-synthetic enzyme **GAD67**. In the provided material this is used operationally to identify the class, not to split it into molecular subclasses. (Seabrook et al., 2013) (Charalambakis et al., 2019)
- **Sox14**
- **Type:** transcript / developmental lineage marker.
- **Context:** precursor and lineage marker for tectum-derived dLGN interneurons during development.
- **Use:** mouse dLGN interneurons arise from **Sox14/Otx2-expressing precursor cells** in the dorsal midbrain. In this setting, Sox14 marks the developmental origin of the class rather than mature synaptic identity inside adult dLGN. (Jager et al., 2016)
- **Otx2**
- **Type:** transcript / developmental lineage marker.
- **Context:** paired with Sox14 as a developmental marker of the precursor population that generates dLGN interneurons.
- **Use:** like Sox14, **Otx2** is informative mainly for embryonic/early postnatal origin and migration into dLGN, not as a demonstrated mature subtype marker in the provided sources. (Jager et al., 2016)
- **mGluR5**
- **Type:** protein expression.
- **Context:** subcellular marker of interneuron **F2 presynaptic dendritic terminals/spines** in retinal triads/glomeruli.
- **Use:** mGluR5 is a notable compartment-specific marker because it is reported on interneuron dendritic spines/F2 profiles close to retinal input and GABA-release machinery. Here it marks a **functional output compartment** of dLGN interneurons rather than the entire cell class, and it is tied to local dendritic GABA release. (Pressler et al., 2013)
- **FGF15-related recruitment program**
- **Type:** transcript-level developmental context, but **not** an interneuron marker per se.
- **Context:** thalamic astrocyte signaling that supports interneuron recruitment into visual thalamus.
- **Use: Fgf15** expression falls when RGC-derived SHH is removed, and this correlates with reduced interneuron recruitment into visual thalamus. This is important for interneuron developmental biology, but it is best understood as an environmental recruitment signal, not a marker of the interneurons themselves. (Somaiya et al., 2022)
- **Parvalbumin (ferret subset)**
- **Type:** protein immunostain.
- **Context:** marker for a **subset** of interlaminar ferret LGNd interneurons with PGN-like properties.
- **Use:** in ferret, a subset of interlaminar interneurons that are electrophysiologically and functionally PGN-like were **parvalbumin-positive**. In anatomical context, this marks a species- and lamina-specific subset near the interlaminar zones, not the main rodent dLGN local interneuron class as a whole. (Sanchez-Vives et al., 1996)

- **Calbindin (ferret context)**
- **Type:** protein immunostain.
- **Context:** mainly useful as a **negative/contrast marker** for PGN-like interlaminar interneurons in ferret LGNd.
- **Use:** calbindin-positive cells were sparse in the interlaminar zones and relatively absent in PGN, while numerous in A and C laminae. So in this context, **lack or rarity of calbindin** helps distinguish the PV-positive PGN-like interlaminar population from other ferret LGN neurons, but calbindin is not presented as a positive marker for dLGN local interneurons. (Sanchez-Vives et al., 1996)
- **What is still missing**
- The provided papers do **not** give a robust mature marker panel comparable to cortical interneuron classes, and they do not provide a well-established transcript-defined taxonomy for rodent dLGN local interneurons.
- The clearest molecular handles in this set are therefore:

1. **GABA/GAD67-related labels** for the class,
2. **Sox14/Otx2** for developmental origin,
3. **mGluR5** for the F2 dendritic output compartment,
4. **parvalbumin** for a ferret PGN-like subset. (Seabrook et al., 2013) (Jager et al., 2016) (Pressler et al., 2013) (Sanchez-Vives et al., 1996)

Evidence

(Somaiya et al., 2022)

Sonic hedgehog-dependent recruitment of GABAergic interneurons into the developing visual thalamus

"Despite these differences, there are some similarities in the cell types of dLGN and vLGN, including the presence and distribution of local interneurons (Arcelli et al., 1997)(Evangelio et al., 2018)(Jaubert-Miazza et al., 2005). In fact, a shared mechanism appears to exist that contributes to the recruitment and integration of these GABAergic neurons into both dLGN and vLGN (Golding et al., 2014;Su et al., 2020)"

".We previously reported that RGC axons signal astrocytes to induce the expression of fibroblast growth factor 15 (FGF15), a motogen required for GABAergic interneuron migration into visual thalamus"

".Specifically, we show that thalamus-projecting RGCs express SHH and thalamic astrocytes generate downstream components of SHH signaling. Deletion of RGC-derived SHH leads to a significant decrease in Fgf15 expression, as well as in the percentage of interneurons recruited into visual thalamus."

(Evangelio et al., 2018)

Thalamocortical Projection Neuron and Interneuron Numbers in the Visual Thalamic Nuclei of the Adult C57BL/6 Mouse

"A key parameter to constrain predictive, bottom-up circuit models of a given brain domain is the number and position of the neuronal populations involved. These include not only the neurons whose bodies reside within the domain, but also the neurons in distant regions that innervate the domain. The mouse visual cortex receives its main subcortical input from the dorsal lateral geniculate nucleus (dLGN) and the lateral posterior (LP) complex of the thalamus. The latter consists of three different nuclei: lateral posterior lateral (LPL), lateral posterior medial rostral (LPMR), and lateral posterior medial caudal (LPMC), each exhibiting specific patterns of connections with the various visual cortical areas. Here, we have determined the number of thalamocortical projection neurons and interneurons in the LP complex and dLGN of the adult C57BL/6 male mouse. We combined Nissl staining and histochemical and immunolabeling methods for consistently delineating nuclei borders, and applied unbiased stereological cell counting methods. Thalamic interneurons were identified using GABA immunolabeling. The C57BL/6 dLGN contains ≈ 21,200 neurons, while LP complex contains ≈ 31,000 total neurons. The dLGN and LP are the only nuclei of the mouse dorsal thalamus containing substantial numbers GABA-immunoreactive interneurons. These interneurons, however, are scarcer than previously estimated; they are 5.6% of dLGN neurons and just 1.9% of the LP neurons. It can be thus inferred that the dLGN contains ≈ 20,000 and the LP complex ≈ 30,400 thalamocortical projection neurons (≈ 12,000 in LPL, 15,200 in LPMR, and 4,200 in LPMC). The present dataset is relevant for constraining models of mouse visual thalamocortical circuits, as well as for quantitative comparisons between genetically modified mouse strains, or across species."

(Seabrook et al., 2013)

Interneurons in the mouse visual thalamus maintain a high degree of retinal convergence throughout postnatal development

"Far less is known about the pattern of retinal convergence onto interneurons, the other principal cell type present in dLGN. These local-circuit neurons are involved in feedforward inhibition onto relay cells and play a role in contrast gain control, shaping receptive fields of relay cells, and altering the temporal precision of retinal inputs (Sherman, 2004)(Wang et al., 2011)"

".Here we overcame these obstacles by using transgenic mice that express enhanced green fluorescent protein (GFP) in γ-aminobutyric acid (GABA)ergic interneurons (GAD67-GFP) (Chattopadhyaya et al., 2004). Such cell-type specific visualization via GFP allowed us to readily target interneurons for in vitro recordings"

"GFP-expressing interneurons were evenly distributed throughout dLGN and had highly complex and widespread dendritic processes that often crossed eye-specific borders. Estimates of retinal convergence derived from excitatory postsynaptic potential (EPSP) amplitude by stimulus intensity plots revealed that unlike relay cells, interneurons recorded throughout the first 5 weeks of life, maintain a large number (approximately eight to ten) of retinal inputs"

".The lack of pruning onto interneurons suggests that the activity-dependent refinement of retinal connections in dLGN is cell-type specific. The high degree of retinal convergence onto interneurons may be necessary for these cells to provide both widespread and local forms of inhibition in dLGN."

(Chattopadhyaya et al., 2004)

Experience and Activity-Dependent Maturation of Perisomatic GABAergic Innervation in Primary Visual Cortex during a Postnatal Critical Period

"The neocortical GABAergic network consists of diverse interneuron cell types that display distinct physiological properties and target their innervations to subcellular compartments of principal neurons. Inhibition directed toward the soma and proximal dendrites is crucial in regulating the output of pyramidal neurons, but the development of perisomatic innervation is poorly understood because of the lack of specific synaptic markers. In the primary visual cortex, for example, it is unknown whether, and to what extent, the formation and maturation of perisomatic synapses are intrinsic to cortical circuits or are regulated by sensory experience. Using bacterial artificial chromosome transgenic mice that label a defined class of perisomatic synapses with green fluorescent protein, here we show that perisomatic innervation developed during a protracted postnatal period after eye opening. Maturation of perisomatic innervation was significantly retarded by visual deprivation during the third, but not the fifth, postnatal week, implicating an important role for sensory input. To examine the role of cortical intrinsic mechanisms, we developed a method to visualize perisomatic synapses from single basket interneurons in cortical organotypic cultures. Characteristic perisomatic synapses formed through a stereotyped process, involving the extension of distinct terminal branches and proliferation of perisomatic boutons. Neuronal spiking in organotypic cultures was necessary for the proliferation of boutons and the extension, but not the maintenance, of terminal branches. Together, our results suggest that although the formation of perisomatic synapses is intrinsic to the cortex, visual experience can influence the maturation and pattern of perisomatic innervation during a postnatal critical period by modulating the level of neural activity within cortical circuits."

(Charalambakis et al., 2019)

Developmental Remodeling of Thalamic Interneurons Requires Retinal Signaling

"little is known about the development of the other principal cell type, intrinsic interneurons"

"in dLGN interneurons"

"their development is a multistaged process involving migration, arbor remodeling, and synapse formation. Arbor remodeling begins during the second postnatal week, after migration to and dispersion within dLGN is complete. This phase includes a period of exuberant branching where arbors grow in number, complexity, and field size. Such growth is followed by branch pruning and stabilization, as interneurons adopt a bipolar architecture"

"To test how such defects influence connectivity with dLGN relay neurons, we used DHPG [(RS)-3,5-dihydroxyphenylglycine], the mGluR1,5 agonist that targets F2 terminals. This led to substantial increases in IPSC activity among WT relay neurons but had little impact in math5−/− mice"

"intrinsic interneurons"

"relies on retinal innervation"

"The absence of retinal signaling disrupts this process. The math5−/− interneurons fail to branch and prune, and instead maintain a simple, sparse architecture"

"GAD67-GFP mice, which express GFP in dLGN interneurons"

(Jager et al., 2016)

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(Pressler et al., 2013)

Metabotropic Glutamate Receptors Drive Global Persistent Inhibition in the Visual Thalamus

"Local interneurons in the dLGN regulate the output of TC cells by releasing GABA from their axonal boutons and specialized dendritic spines"

".These local interneurons have long been of interest because, in additional to conventional axodendritic synapses, they inhibit TC cells via specialized dendrodendritic synapses that involve the convergence of an RGC axon, an interneuron dendritic spine, and a relay cell dendrite, collectively known as a triad (Rafols and Valverde, 1973;Ohara et al., 1983;(Ohara et al., 1983). RGC inputs to the interneuron spine are in close proximity to GABA release machinery and to metabotropic glutamate receptors type 5 (mGluR5s) expressed in interneurons (Ohara et al., 1983). This, along with the observation that mGluR5 agonists evoke GABA release from interneuron dendrites in the absence of spiking (Cox et al., 1998;Cox and Sherman, 2000;Govindaiah and Cox, 2004;(Ohara et al., 1983), suggested that RGC axons trigger local GABA release by activating mGluR5 on interneuron spines and locally elevating calcium (Netzeband et al., 1997;Tempia et al., 1998)."

(Sanchez-Vives et al., 1996)

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"The ferret dorsal lateral geniculate nucleus (LGNd) contains interneurons within the interlaminar zones situated between the laminae corresponding to the ipsi-and contralateral eyes. We found that a subset of these neurons exhibits electrophysiological properties similar to those previously reported for perigeniculate (PGN) neurons, including the generation of rhythmic sequences of rebound low-threshold Ca 2 spikes at a frequency of 1-4 Hz after the intracellular injection of a hyperpolarizing current pulse. These "PGN-like" interlaminar interneurons innervated restricted regions of the A-laminae, inhibited thalamocortical cells through GABA A , and perhaps GABA B , receptors, and were excited by axon collaterals from thalamocortical cells. This reciprocal relationship is identical to that formed by PGN cells and allowed the PGN-like interlaminar neurons to participate in the generation of spindle waves and other network oscillations. Pharmacologically, PGN-like interlaminar interneurons were also similar to PGN neurons: both generated a prolonged depolarization in response to the local application of serotonin, 1S,3R-ACPD, and CCK8S, and a rapid depolarization followed by a more prolonged hyperpolarization in response to acetylcholine"

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Transcriptomics and related molecular datasets

Transcriptomic coverage of dLGN local GABAergic interneurons is still limited compared with cortex. The clearest molecular datasets so far support the idea that mouse dLGN interneurons form one main tectum-derived class, with some evidence for modest within-class diversity rather than many sharply separated transcriptomic types. (LLM Memory)

- **Single-cell / single-nucleus atlases of mouse thalamus**
- Broad mouse thalamus transcriptomic atlases include inhibitory cells from sensory thalamus and can recover dLGN interneuron-like populations, usually through **Gad1/Gad2**, **Sox14**, and related GABAergic signatures.
- In these datasets, dLGN local interneurons tend to group closer to other visual-thalamic interneurons than to TRN neurons, which fits their distinct tectal lineage.

- A repeated theme from these atlas-scale studies is that dLGN interneurons do **not** split into a large number of stable molecular classes the way cortical interneurons do; instead they often appear as one main cluster or a small number of nearby clusters. (Model-Generated)
- **Lineage-linked transcriptomic evidence**
- Work on developmental origin gives an important molecular handle even when full adult subtype taxonomy is limited: dLGN interneurons derive from **Sox14/Otx2-positive** dorsal midbrain precursors.
- In transcriptomic context, this lineage is useful because it helps separate local dLGN interneurons from other thalamic inhibitory neurons, especially TRN cells, which have a different origin.
- This is best viewed as a **developmental molecular identity** for the class, not yet as a mature adult subtype scheme. (Model-Generated)
- **Evidence for limited heterogeneity from combined physiology and molecular profiling**
- A mouse study that combined physiology with molecular readouts reported two interneuron groups, often described as **small-cell** and **large-cell** classes.
- The main molecular feature reported there was **nNOS/Nos1**, seen in the larger-diameter group and not in the smaller group.
- This supports the idea that if heterogeneity exists, it may be shallow and linked to electrophysiology and cell size rather than to many deeply separated transcriptomic branches. (Model-Generated)
- **Patch-seq / project-seq / spatial transcriptomics**
- I am not confident that there is yet a widely used, dLGN-interneuron-focused **Patch-seq**, **project-seq**, or **spatial transcriptomics** dataset that cleanly resolves multiple local interneuron subtypes and is standard in the field.
- Related work exists at the level of connectomics, physiology, and broad single-cell atlases, but a subtype-resolved molecular resource for dLGN local interneurons is still much thinner than for cortical interneurons. (Model-Generated)
- **What current transcriptomic data seem to say about cell type status**
- Current molecular data fit the older anatomical view that rodent dLGN local interneurons are a relatively compact cell class.
- If subtypes are present, the best-supported split at present is a **small number of states or subclasses**, not a large taxonomy.
- Candidate axes of variation include **Nos1/nNOS status**, cell size, intrinsic excitability, and perhaps differences tied to dendritic output mode or retinal/corticothalamic sampling, but these links are still not fully mapped at transcriptome scale. (Model-Generated)
- **Dataset accession**
- I am not sure enough of specific accession numbers for dLGN-interneuron-resolved transcriptomic datasets to list them reliably here.
- If you want, I can next help assemble a cautious table of likely datasets and accessions to verify manually before inclusion. (Model-Generated)